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(54) **HEAT TRANSFER**

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(2013.01); **F02C 7/16** (2013.01); **F02C 7/32**
(2013.01); **F23R 3/005** (2013.01); **F05D**
2260/20 (2013.01)

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2005/0155353 A1 7/2005 Sabatino et al.
2011/0230686 A1 9/2011 Rusek et al.
(Continued)

FOREIGN PATENT DOCUMENTS

EP 3179075 A1 6/2017
JP 2007-285122 A 11/2007
WO 2013/165487 A2 11/2013

OTHER PUBLICATIONS

Silitonga, A.S., et. al., Overview properties of biodiesel diesel
blends from edible and non-edible feedstock, 2013, Elsevier, Renew-
able and Sustainable Energy Reviews, pp. 346-360 (Year: 2013).
(Continued)

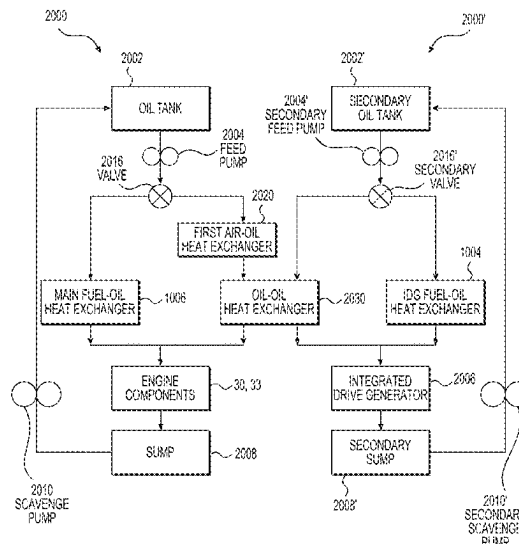
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(57) **ABSTRACT**

A method of operating a gas turbine engine having an engine
core having a turbine, a compressor, a combustor arranged to
combust a fuel, and a core shaft connecting the turbine to
the compressor; a fan located upstream of the engine core;
a fan shaft; a gearbox that receives an input from the core
shaft and outputs drive to the fan via the fan shaft; a heat
exchange system having at least one fuel-oil heat exchanger
arranged to transfer heat to the fuel; and a fuel pump
arranged to deliver the fuel to the combustor, wherein the
fuel pump is located downstream of the at least one fuel-oil
heat exchanger. The method comprises controlling the heat
exchange system so as to raise the fuel temperature to at
least 135° C. on entry to the combustor at cruise conditions.

20 Claims, 12 Drawing Sheets



- (51) **Int. Cl.**
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(56) **References Cited**

U.S. PATENT DOCUMENTS

2011/0232293	A1	9/2011	Leese
2012/0238787	A1	9/2012	Gruber et al.
2013/0086909	A1	4/2013	Wang et al.
2013/0087308	A1	4/2013	Logan et al.
2016/0024968	A1	1/2016	Stearns et al.
2019/0170016	A1	6/2019	Stearns et al.
2019/0277201	A1*	9/2019	Veilleux, Jr. B64D 37/32
2023/0332543	A1	10/2023	Bemment et al.

OTHER PUBLICATIONS

Freedman, Andrew, A milestone for more sustainable air travel, *Axios*, Dec. 2, 2021, accessed at <https://www.axios.com/2021/12/02/united-airlines-sustainable-aviation-fuels> Jan. 12, 2024 (Year: 2021).

Appeldoorn, J.K., et. al., Lubricity Properties of High-Temperature Jet Fuels, Esso Research and Engineering Co., Quarterly Report No. 9, 1967 (Year: 1967).

Kuszewski, H., et. al., Effect of Temperature Tribological Properties of 1-Butanol-Diesel Fuel Blends—Preliminary Experimental Study Using the HFFR Method, *El Sevier*, 2021 (Year: 2021).

Appeldoorn, J.K., et. al., Lubricity Properties of High-Temperature Jet Fuels, Esso Research and Engineering Co., Technical Report AFAPL-TR-66-89, Part 1, Aug. 1966 (Year: 1966).

Jaaskelainen, H., Fuel Property Testing: Lubricity, DieselNet, 2008, accessed at https://dieselnet.com/tech/fuel_diesel_lubricity.php on Mar. 22, 2024 (Year: 2008).

Sivebaek, I.M., et. al., The Lubricity of Diethyl Ether (DEE), Proceedings of 16th Nordic Symposium on Tribology, 2014, accessed at https://backend.orbit.dtu.dk/ws/portalfiles/portal/101032575/NT2014_120.pdf Mar. 22, 2024 (Year: 2014).

Lacey, P.I., Effect of Fuel Composition and Prestressing on Lubricity, Interim Report, U.S. Army TARDEC, 1995, accessed at <https://apps.dtic.mil/sti/tr/pdf/ADA297747.pdf> on Mar. 22, 2024 (Year: 1995).

May 10, 2024 Search Report issued in British Patent Application No. GB2319067.1.

Jig, “Aviation Fuel Quality Requirements for Jointly Operated Systems,” Product Specifications Bulletin, Bulletin No. 141, AFQRJOS Checklist Issue 33, Jun. 4, 2022, pp. 1-10.

May 13, 2024 European Search Report issued in European Patent Application No. 23216510.0.

Rolls-Royce, “Rolls-Royce conducts first tests of 100% sustainable aviation fuel for use in business jets”, Feb. 1, 2021, accessed at <https://www.rolls-royce.com/media/press-releases/2021/01-02-2021-business-aviation-rr-conducts-first-tests-of-100-percent-sustainable-aviation-fuel.aspx> Feb. 27, 2024 (Year: 2021).

United, “United to Become First in Aviation History to Fly Aircraft Full of Passengers Using 100% Sustainable Fuel”, Dec. 1, 2021, accessed at <https://www.united.com/en/us/newsroom/announcements/united-to-become-first-in-aviation-history-to-fly-aircraft-full-of-passengers-using-100-sustainable-fuel> (Year: 2021).

Snijders et al., “Impact of Fuel Composition on Emissions and Performance of GTL Kerosine Blends in a Cessna Citation II,” *IASH 2011*, the 12th International Conference on Stability, Handling and Use of Liquid Fuels, Oct. 1, 2011, pp. 16-20.

Yang et al., “An overview on performance characteristics of bio-jet fuels,” *Fuel*, IPC Science and Technology Press, vol. 237, Oct. 16, 2018, pp. 916-936.

* cited by examiner

FIG. 2

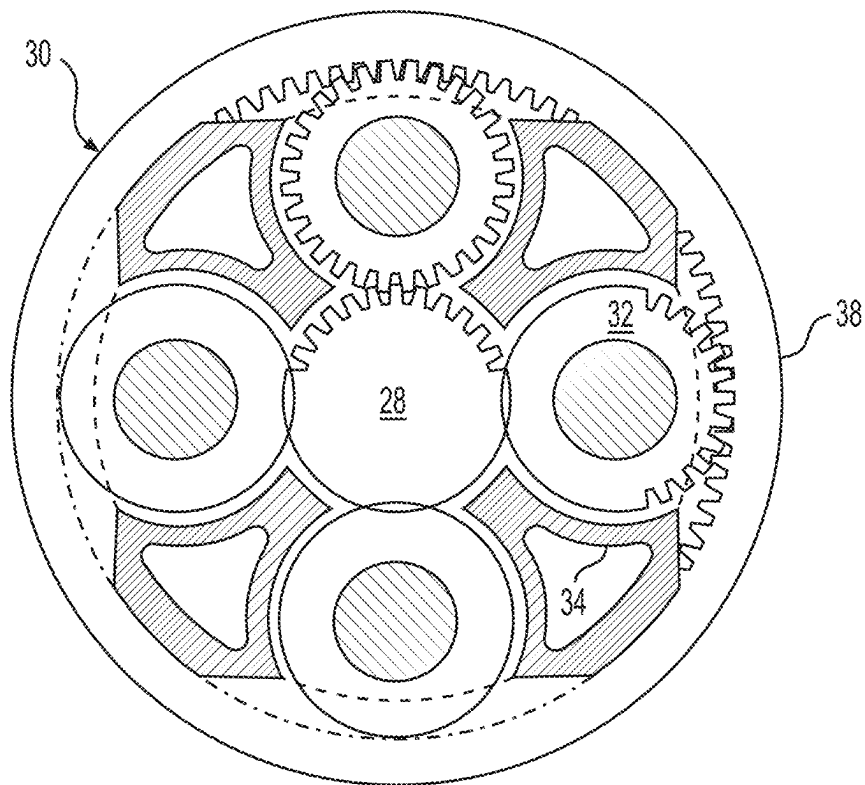


FIG. 3

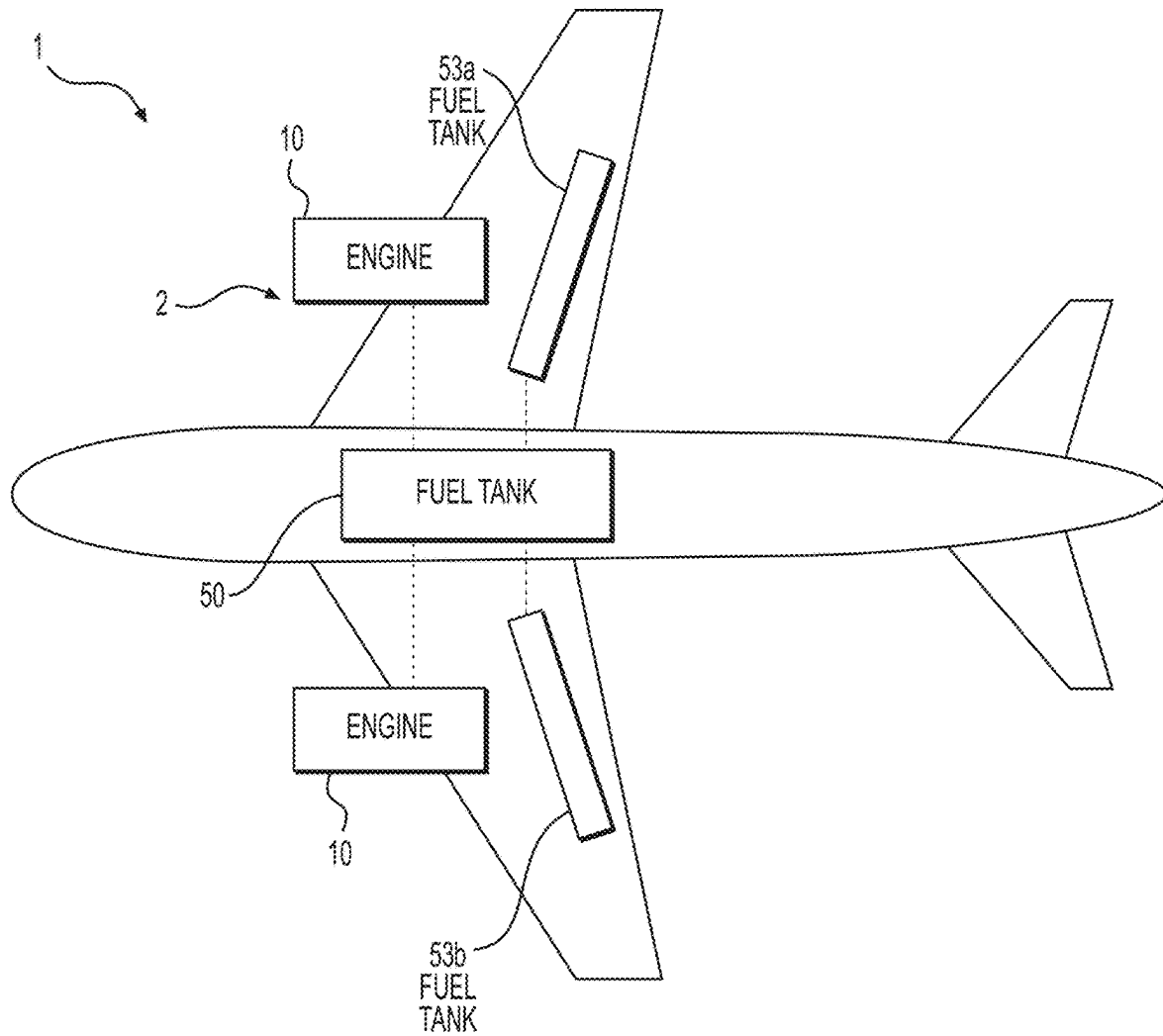


FIG. 4

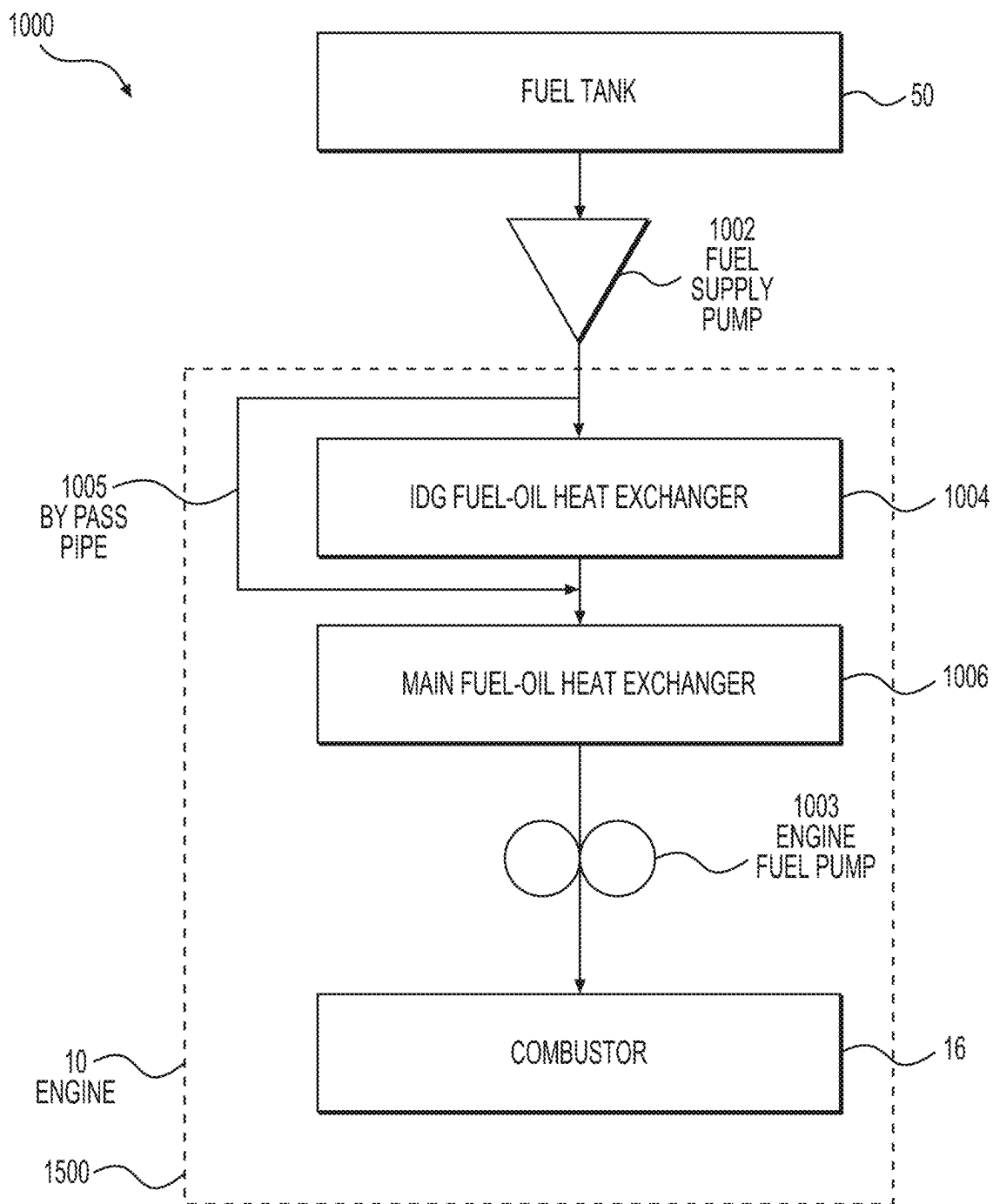


FIG. 5

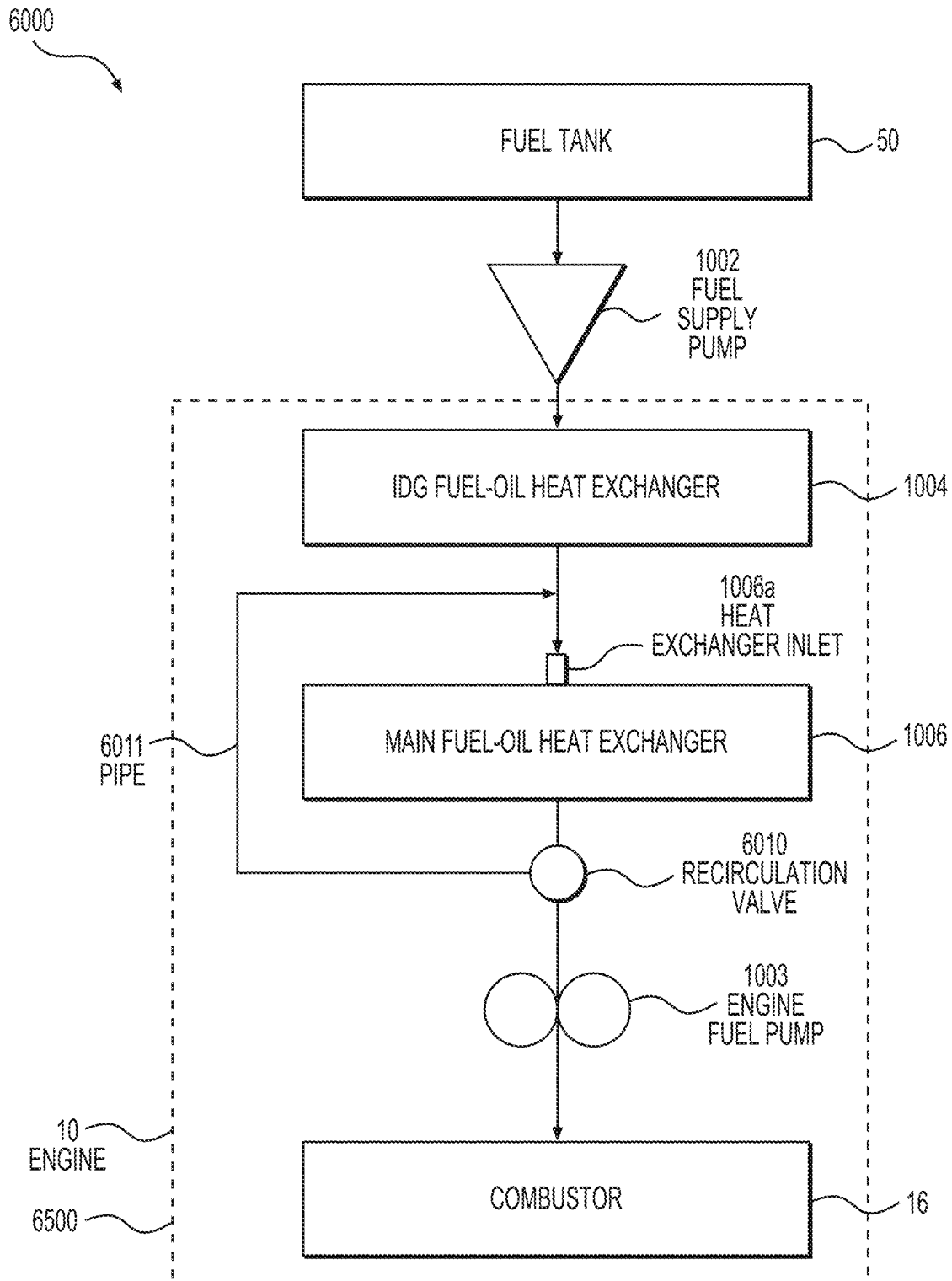


FIG. 6

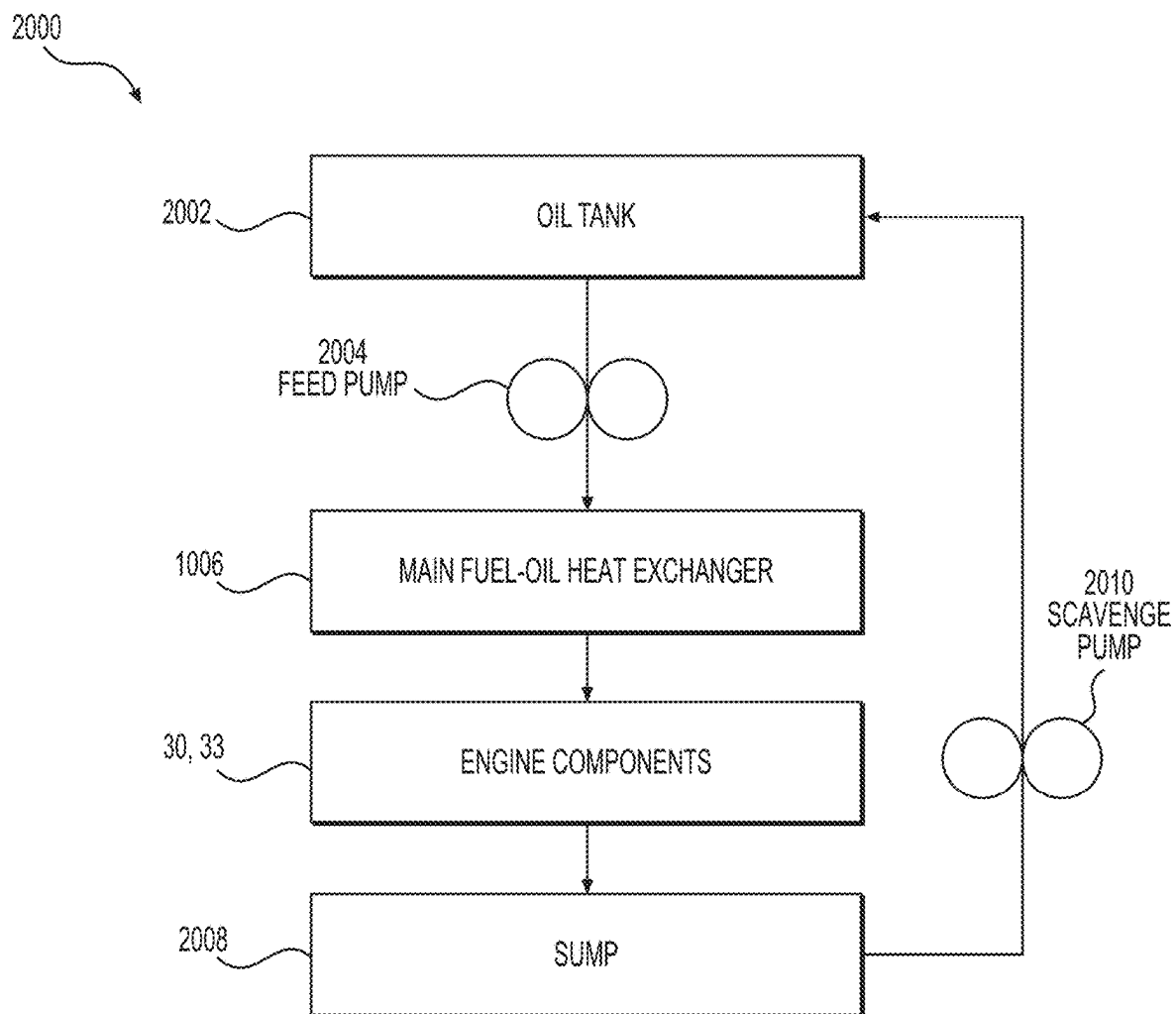


FIG. 7

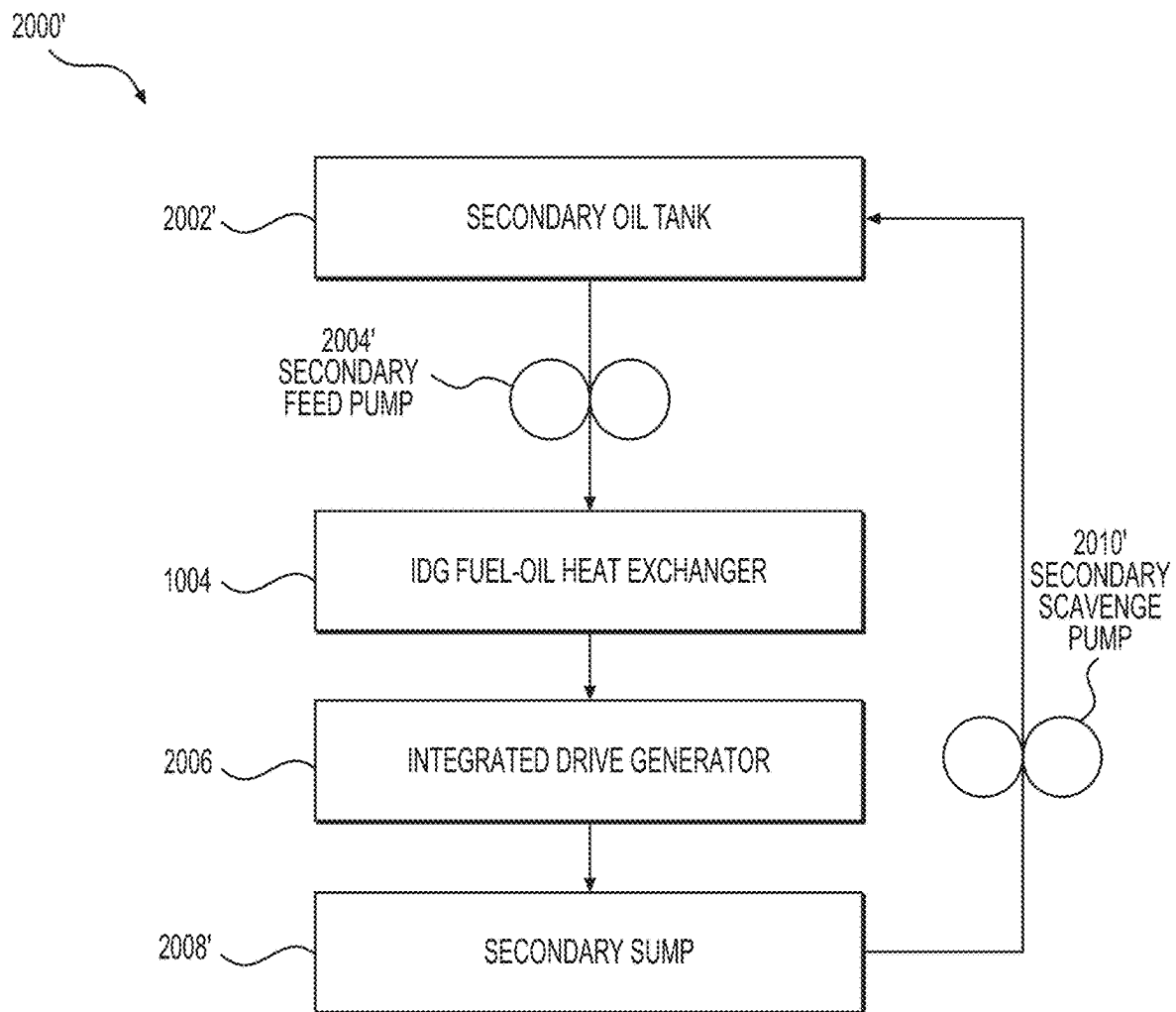


FIG. 8

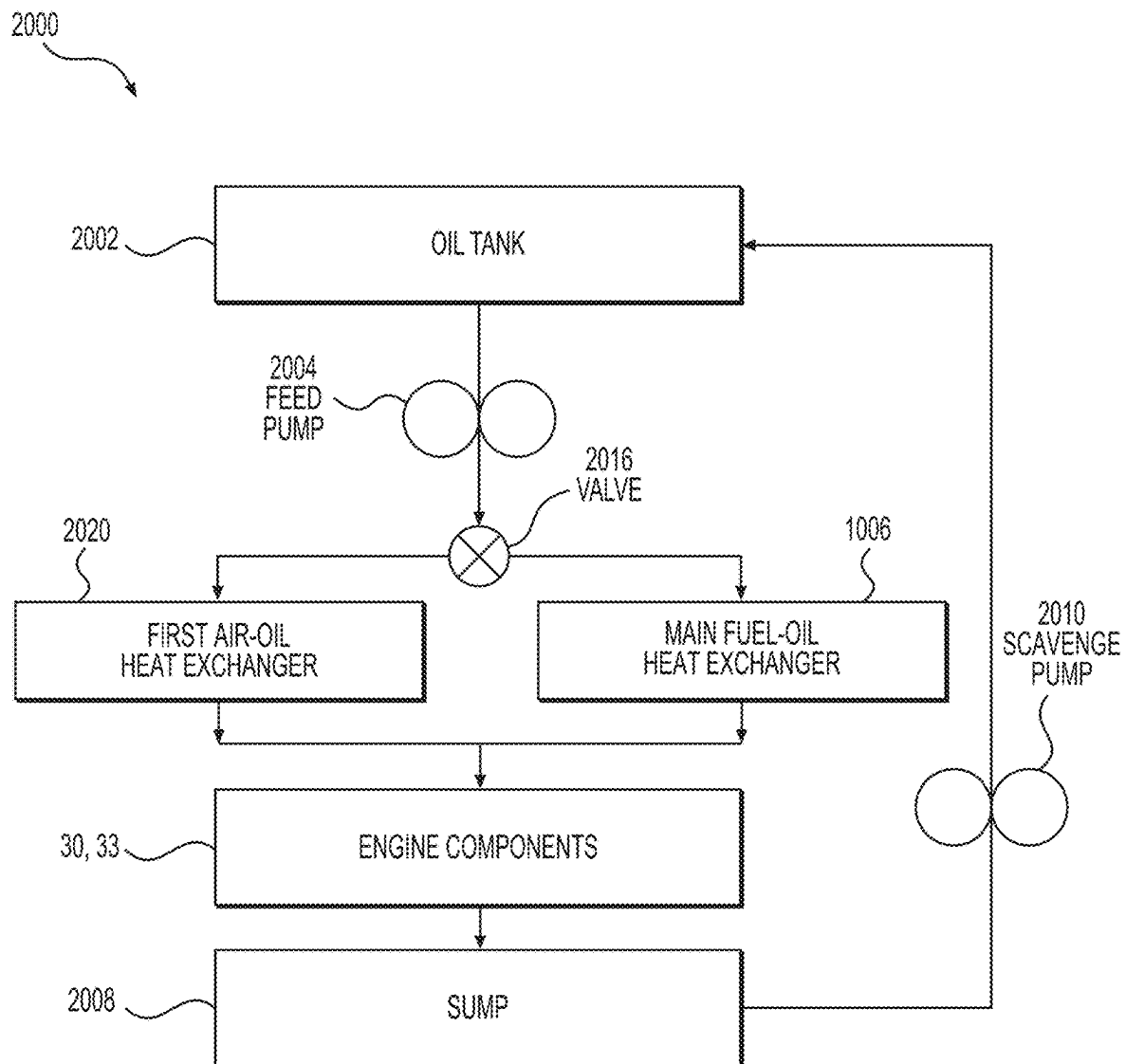


FIG. 9

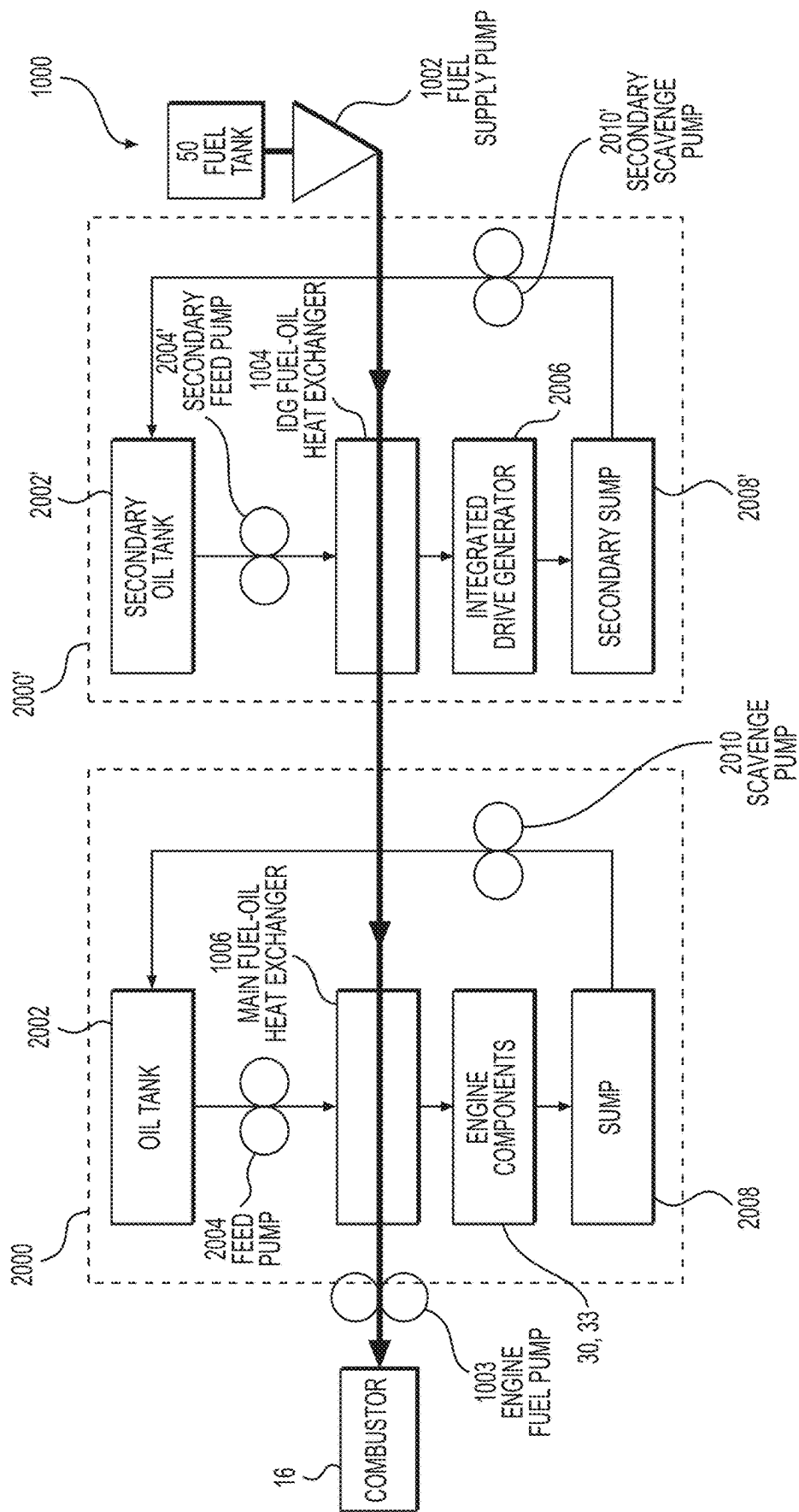


FIG. 10

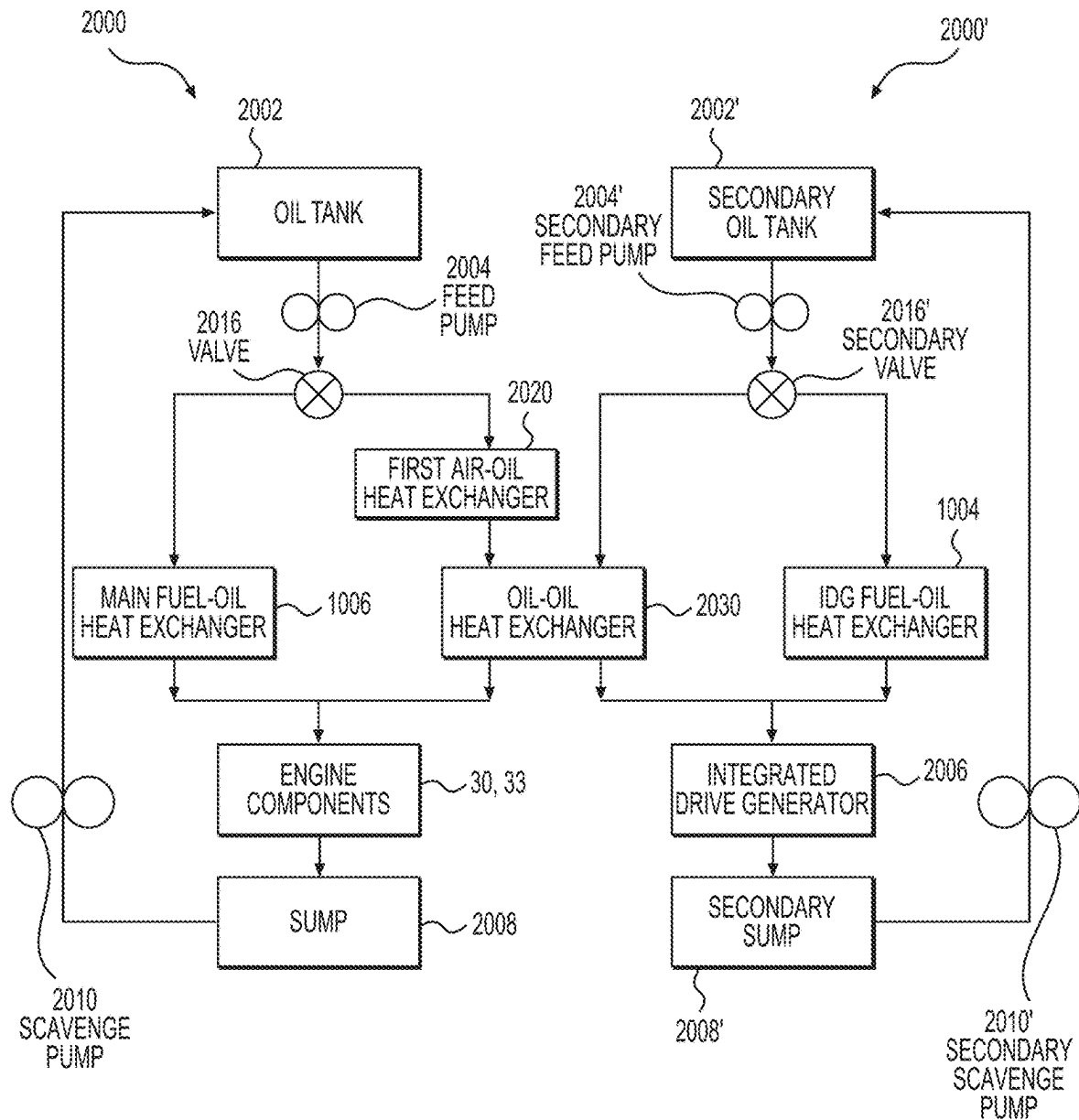


FIG. 11

12000

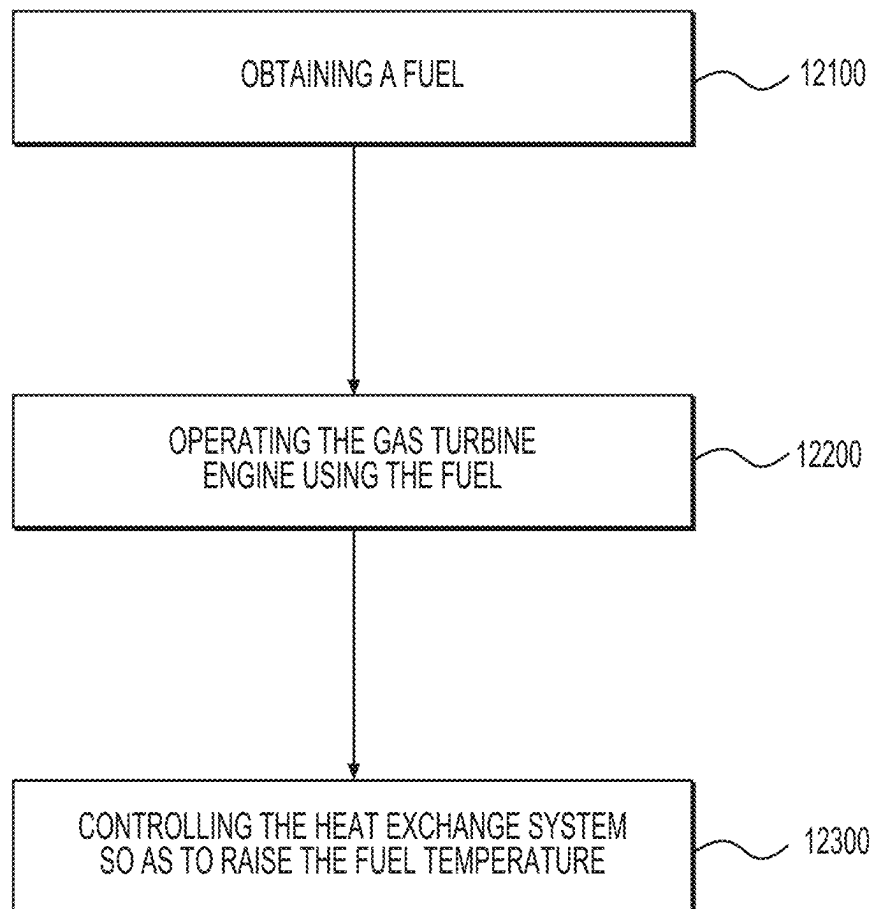


FIG. 12

13000

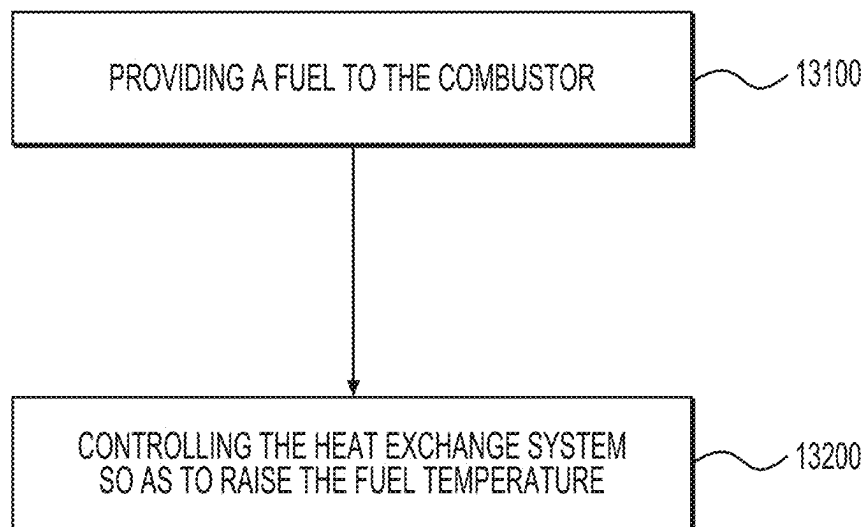
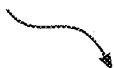


FIG. 13

1

HEAT TRANSFER

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/337,497, filed on 20 Jun. 2023, which claims priority from United Kingdom Patent Application Number 2219411.2 filed on 21 Dec. 2022. The entire contents of each of these applications are incorporated herein by reference in their entirety.

BACKGROUND

Field of the Disclosure

The present disclosure relates to aircraft propulsion systems, and to methods of operating aircraft involving the management of different fluids and heat transfer therebetween, and in particular to management of fuel properties on entry to the combustor.

Description of the Related Art

There is an expectation in the aviation industry of a trend towards the use of fuels different from the traditional kerosene-based jet fuels generally used at present. The fuels may have differing fuel characteristics relative to petroleum-based hydrocarbon fuels.

Thus, there is a need to take account of fuel properties for these new fuels and to adjust the methods of operating gas turbine engines.

SUMMARY

According to a first aspect there is provided a method of operating a gas turbine engine, the gas turbine engine comprising:

- an engine core comprising a turbine, a compressor, a combustor arranged to combust a fuel, and a core shaft connecting the turbine to the compressor;
 - a fan located upstream of the engine core;
 - a fan shaft;
 - a gearbox that receives an input from the core shaft and outputs drive to the fan via the fan shaft;
 - a heat exchange system comprising one or more heat exchangers arranged to transfer heat to the fuel, the one or more heat exchangers comprising at least one fuel-oil heat exchanger; and
 - a fuel pump arranged to deliver the fuel to the combustor, wherein the fuel pump is located downstream of the at least one heat exchanger;
- wherein the method comprises controlling the heat exchange system so as to raise the fuel temperature to at least 135° C. on entry to the combustor at cruise conditions.

The inventors have appreciated that the use of fuels different from the traditional kerosene-based jet fuels, such as sustainable aviation fuels, may result in different fuel properties, and that parameters at cruise conditions may be adjusted to make use of the different fuel properties. In particular, some fuels may be heated to higher temperatures in one or more fuel-oil heat exchangers than traditional fuels. This may improve combustion efficiency of the fuel. For example, a higher-temperature fuel may be less viscous which may result in a different nozzle spray-pattern and droplet size within the combustor, so changing fuel-air

2

mixing and improving combustion efficiency. Higher fuel temperatures on entry to the combustor may also allow for a method providing improved oil cooling (as the fuel is able to take more heat). Heat transfer may be optimised to tailor the fuel spray characteristics within the combustor.

The fuel flow through the one or more heat exchangers, and/or one or more other properties of the heat exchange system, may be controlled so as to raise the fuel temperature to between 135° C. and 170° C. on entry to the combustor at cruise conditions.

The fuel flow may be controlled so as to raise the fuel temperature to between 150° C. and 170° C. on entry to the combustor at cruise conditions.

The fuel temperature may be raised to at least 150° C., 170° C., or 190° C. on entry to the combustor at cruise conditions.

The fuel flow may be controlled so as to raise the fuel temperature to between 135-150° C., 135-160° C., 135-170° C., 135-180° C., 135-190° C., or 135-200° C. on entry to the combustor at cruise conditions. The fuel flow may be controlled so as to raise the fuel temperature to an average of at least 140° C., 150° C., 160° C., 170° C., 180° C., 190° C., or 200° C. on entry to the combustor at cruise conditions.

The fuel flow may be controlled so as to raise the fuel temperature to a maximum of 170° C., 180° C., 190° C., or 200° C. on entry to the combustor at cruise conditions. The setting of a maximum temperature may serve to protect the fuel pump and/or other engine components from excessively high temperatures which could decrease its lifespan.

The fuel pump may be located downstream of all heat exchangers of the heat exchange system.

The fuel temperatures on entry to the combustor at cruise conditions may be defined as an average over at least 5 minutes, and optionally over ten minutes, under steady state cruise conditions. These average temperatures do not include transient spikes in temperature, which may be defined as fluctuations in temperature of the fuel during operation, often temporary elevation of the temperature. Each fluctuation may not last more than 5 minutes. A fuel temperature of at least 135° C. on entry to the combustor at cruise conditions therefore requires the fuel temperature to remain at or above 135° C. over a period of time, and a transient spike to a temperature above 135° C. is not sufficient.

The heat exchange system may comprise an air-oil heat exchanger arranged to transfer heat from the oil flowing therethrough to the air. The air-oil heat exchanger and a fuel-oil heat exchanger may be arranged in a parallel configuration on different branches of an oil loop system. The method may comprise adjusting the amount of oil sent to the air-oil heat exchanger based on fuel temperature.

The controlling the heat exchange system so as to raise the fuel temperature may therefore comprise controlling oil flow through one or more heat exchangers.

In implementations with two fuel-oil heat exchangers—a primary fuel-oil heat exchanger and a secondary fuel-oil heat exchanger—the controlling the heat exchange system so as to raise the fuel temperature may additionally or alternatively comprise adjusting an amount of fuel sent through the primary fuel-oil heat exchanger and the secondary fuel-oil heat exchanger. This adjustment may comprise adjusting what proportion of fuel is sent down a bypass pipe passing the heat exchanger, rather than being sent through the heat exchanger. Alternatively or additionally, this adjustment may comprise adjusting what proportion of fuel is recirculated from an outlet of a heat exchanger back to an inlet of that heat exchanger. The primary fuel-oil heat

3

exchanger and the secondary fuel-oil heat exchanger may be located on separate oil loops—for example a primary oil loop arranged to lubricate and cool a gearbox of the engine and a secondary oil loop arranged to lubricate and cool an integrated drive generator of the engine, as described in more detail below.

The heat exchange system may therefore comprise at least one bypass pipe arranged to allow fuel or oil to bypass a heat exchanger of the heat exchange system. The method may comprise adjusting the amount of fuel or oil sent through the bypass pipe based on fuel temperature. In some implementations, the or each heat exchanger may comprise at least one bypass pipe, and may comprise two bypass pipes (one each for fuel and oil).

One or more valves may be actuated to perform the adjustment of oil flow and/or of fuel flow. One or more temperature sensors may be used to provide feedback for automatic adjustment, controlled by a controller of the heat exchange system, which may be a dedicated controller or a part of the engine electronic controller (EEC).

According to a second aspect, there is provided a gas turbine engine for an aircraft comprising:

- an engine core comprising a turbine, a compressor, a combustor arranged to combust a fuel, and a core shaft connecting the turbine to the compressor;
 - a fan located upstream of the engine core;
 - a fan shaft;
 - a gearbox arranged to receive an input from the core shaft and outputs drive to the fan via the fan shaft;
 - a heat exchange system comprising at least one fuel-oil heat exchanger arranged to transfer heat to the fuel; and
 - a fuel pump arranged to deliver the fuel to the combustor, wherein the fuel pump is located downstream of the at least one fuel-oil heat exchanger;
- wherein the heat exchange system is arranged to heat the fuel so as to raise the fuel temperature to at least 135° C. on entry to the combustor at cruise conditions.

The gas turbine engine may further comprise an accessory gearbox and a pump shaft, wherein the accessory gearbox receives an input from the core shaft and outputs drive to the fuel pump via the pump shaft. The pump speed may therefore be linked to core shaft rotation speed, and may have, by means of the gearbox, a fixed speed relationship between the core shaft (or in some engine architectures, another intermediate shaft) and the fuel pump. In some implementations, the accessory gearbox may instead be arranged to provide a fixed number of different speeds for a given core shaft speed. The recirculation of fuel may allow a wider variety of fuel flow rates into the combustor for the given number of pump speeds available at a certain shaft speed. The recirculation of fuel may allow multiple different fuel flow rates into the combustor at a given core shaft speed in implementations in which there is a single fixed speed relationship between the core shaft and the fuel pump.

It is envisaged that all features described for the first aspect can equally apply to the second aspect.

According to a third aspect, there is provided a method of operating a gas turbine engine, the gas turbine engine comprising:

- an engine core comprising a turbine, a compressor, a combustor arranged to combust a fuel, and a core shaft connecting the turbine to the compressor;
- a fan located upstream of the engine core;
- a fan shaft;
- a gearbox that receives an input from the core shaft and outputs drive to the fan via the fan shaft;

4

- a heat exchange system comprising one or more heat exchangers arranged to transfer heat to the fuel, the one or more heat exchangers comprising at least one fuel-oil heat exchanger; and

- a fuel pump arranged to deliver the fuel to the combustor, wherein the fuel pump is located downstream of the at least one fuel-oil heat exchanger;

wherein the method comprises operating the gas turbine engine using a fuel having a lubricity of between 0.71 mm and 0.90 mm wear scar diameter (WSD) at 25° C.

Using a fuel having the desired lubricity may comprise selection of a suitable fuel. The selection of the fuel may comprise selecting a single fuel. The selection of the fuel may comprise selecting a fuel mixture. The selected fuel may be the only fuel on board of the aircraft. As such, the selection of the fuel may be performed during refuelling of the aircraft. Alternatively, the selected fuel may be one of several fuels on board the aircraft, or a mixture of multiple fuels on board the aircraft; the selection may therefore be performed in flight.

In order to obtain a desired lubricity on entry to the pump or combustor at cruise conditions, the fuel may also be heated using the heat exchange system. The method of operation of the engine may therefore comprise a combination of selection of a suitable fuel and suitable heating of the fuel during use, so as to obtain a desired lubricity at cruise conditions, in particular on entry to the pump and on entry to the combustor. The control of the heat exchange system may be adjusted depending on the fuel. The adjustment may be as described with respect to the first and second aspects.

Fuel leaving the fuel pump may be fed directly to the combustor—the fuel temperature on entry to the fuel pump may therefore be at least substantially equal to the fuel temperature on entry to the combustor, and the fuel lubricity at these two positions may also be at least substantially equal.

Lubricity is measured by one or more of a series of controlled experiments. One such test involves placing a hardened steel ball against another hardened steel object (e.g. a disc or a cylinder, depending on the selected test type) with a load applied, to create a force between the two. The fuel sample being tested is poured into a cup that contains these steel samples, and a motor is turned on to force the ball to rub back and forth against the steel disk or cylinder at a fixed rate for a set period of time. The ball is then removed and the width and length of the “wear scar” (i.e. the mark on the ball caused by the wear against the steel disk) is measured. The average of these two dimensions is known as the wear scar diameter, or WSD. The smaller the scar, the less wear has occurred and so the higher the lubricity of the fuel tested.

The Ball-on-Cylinder Lubricity Evaluator (BOCLE) test was specifically developed for aviation jet fuels as it is particularly useful for measuring the effects of fuels and additives on oxidative wear, which is an important mechanism in aircraft fuel systems. For a BOCLE test, a fixed steel ball and a rotating metal cylinder are used, submerged in a sample of the fluid to be tested (e.g. fuel). A load is applied to press the ball against the cylinder whilst the cylinder is rotated, to simulate metal-metal contact in operation and assess the effect of the sample fluid. The presence and nature of “wear scars” on the ball following the test allow lubrication properties of the fluid sample at the control speeds and contacting loads to be assessed.

For the WSD values provided herein, the ASTM D5001 BOCLE test was used, with a temperature of 25° C.±1° C., and a duration of 30 minutes±1 minute.

5

In this standard test, the ball used is made of a chrome alloy steel that is AISI standard steel No. E-52100 and has a diameter of 12.7 mm. The metal cylinder is made of SAE 8720 steel. The steel ball is pressed onto the metal cylinder with an applied load of 1000 g, which remains constant throughout. The metal cylinder is mounted to a motor which rotates it at a fixed velocity of 240 rotations per minute for thirty minutes. Once the test has been completed, the components are disassembled, and the steel ball is examined using a microscope at 100x zoom to determine the wear scar diameter of the wear scar formed during the test.

That standard temperature and time period were used for the values listed herein, using the testing protocol as laid out above, unless otherwise specified.

Positioning the pump downstream of the primary fuel-oil heat exchanger may allow the performance of the pump to be improved by the changed lubricity of the fuel after heating. Traditional fuels may have an unsuitable lubricity for optimal pump performance after the desired level of heating. The different fuels, such as sustainable aviation fuels, may not have a sufficiently high lubricity for use with the fuel pump before heating by the one or more heat exchangers, hence the fuel pump being positioned downstream of the one or more heat exchangers. Combining novel fuel characteristics with the increased heating of said fuel can provide a balance of parameters to achieve a desired lubricity. Lubricity of the fuel can also affect combustion efficiency, in particular with respect to fuel nozzle spray performance within the combustor. Fuel nozzle spray performance affects burn efficiency of the fuel. Engine efficiency can therefore be improved with the selection of desired lubricity. For example, changes in lubricity of the fuel may result in a different nozzle spray-pattern and droplet size within the combustor, so changing fuel-air mixing; tailoring of lubricity may therefore be used to improve combustion efficiency. Heat transfer may be optimised to tailor the fuel spray characteristics within the combustor. Lubricity of the fuel provides a measure of how much wear surfaces lubricated by the fuel will suffer—a more lubricious fuel will reduce friction, so protecting surfaces (such as inner surfaces of pumps or valves) from wear. A low fuel lubricity may cause increased wear or scarring in pumps and injectors, for example. Lubricity may therefore be tailored to lie in a specific range for improved engine performance and/or longevity.

A fuel having a lubricity of between 0.71 mm and 0.85 mm WSD or between 0.75 mm and 0.90 mm WSD may be used. A fuel having a lubricity of between 0.80 mm and 0.90 mm WSD may be used. A fuel having a lubricity of between 0.80 mm and 0.85 mm WSD may be used. A fuel having a lubricity of between 0.85 mm and 0.90 mm WSD may be used.

A fuel having a lubricity of greater than 0.75 mm WSD may be used. A fuel having a lubricity of greater than 0.80 mm WSD may be used. A fuel having a lubricity of greater than 0.85 mm WSD may be used. A fuel having a lubricity of greater than or equal to 0.90 mm WSD may be used.

The intrinsic lubricity of a selected fuel may be lower than traditional kerosene fuels. However, the heating of the fuel to a higher temperature compared to that with traditional fuels may compensate—heating the fuel to an elevated temperature can increase the lubricity (i.e. reduce the WSD value) to a desired value.

For example, operating the gas turbine engine may comprise controlling the heat exchange system so as to raise the fuel temperature to at least 135° C. on entry to the combustor at cruise conditions. The fuel flow through the one or more

6

heat exchangers, and/or one or more other properties of the heat exchange system, may be controlled so as to raise the fuel temperature. The fuel temperature may be raised to at least 150° C., 170° C., or 190° C. on entry to the combustor at cruise conditions.

According to a fourth aspect, there is provided a gas turbine engine for an aircraft comprising:

- an engine core comprising a turbine, a compressor, a combustor arranged to combust a fuel, and a core shaft connecting the turbine to the compressor;
 - a fan located upstream of the engine core;
 - a fan shaft;
 - a gearbox arranged to receive an input from the core shaft and outputs drive to the fan via the fan shaft;
 - a heat exchange system comprising at least one fuel-oil heat exchanger arranged to transfer heat to the fuel; and
 - a fuel pump arranged to deliver the fuel to the combustor, wherein the fuel pump is located downstream of the at least one fuel-oil heat exchanger;
- wherein the fuel has a lubricity of between 0.71 mm and 0.90 mm wear scar diameter (WSD) at 25° C.

It is envisaged that all features described for the third aspect can equally apply to the fourth aspect.

Further, the third and/or fourth aspects may be implemented in conjunction with the first and/or second aspects.

In any of the above aspects, the gas turbine engine may comprise a primary oil loop system arranged to supply oil to the gearbox; and the one or more heat exchangers of the heat exchange system may comprise at least a primary fuel-oil heat exchanger through which the oil of the primary oil loop system and fuel flows to transfer heat from the oil to the fuel.

Using the fuel to take more heat from the oil, rather than relying on heat transfer from the oil to the environment/air (e.g., in an oil-air heat exchanger) provides a more thermally efficient engine. This may improve the cooling of the oil before it is returned to where it is used for lubrication and/or cooling in the turbine engine. Additionally, the improved cooling of the oil may in turn improve the cooling effect of the oil on the components of the engine through which it flows.

The gas turbine engine may further comprise:

- an integrated drive generator; and
- a secondary oil loop system arranged to provide oil to the integrated drive generator.

The heat exchange system may comprise an oil-oil heat exchanger arranged to transfer heat between the oil of the primary loop system and the oil of the secondary loop system.

The oil of the primary oil loop system may be physically separate from the oil of the secondary oil loop system, as such the primary and secondary oil loop systems are distinct, separate closed loops. As such, the primary oil loop system may be referred to as a primary closed loop oil system and the secondary oil loop system may be referred to as a secondary closed loop oil system. The oil of the primary oil loop system may or may not be chemically similar or the same as the oil of the secondary oil loop system. That is to say that the primary and secondary oil loop systems may be fluidly isolated such that oil never mixes between the two, and the oil in each may be chemically distinct and/or may be pumped at a different rate.

The one or more heat exchangers of the heat exchange system may further comprise a secondary fuel-oil heat exchanger arranged to receive the fuel and oil from the secondary oil loop system. Heat may be transferred between

the oil from the secondary oil loop system and the fuel using the secondary fuel-oil heat exchanger.

As such, the gas turbine engine may further comprise an integrated drive generator; and a secondary oil loop system arranged to provide oil to the integrated drive generator; and the at least one fuel-oil heat exchanger may further comprise a secondary fuel-oil heat exchanger arranged such that oil from the secondary oil loop system and the fuel flows therethrough, and further arranged to transfer heat between the oil from the secondary oil loop system and the fuel.

The fuel may flow through the secondary fuel-oil heat exchanger prior to flowing through the primary fuel-oil heat exchanger, such that heat is transferred from the oil in the secondary oil loop system to the fuel before heat is transferred from the oil in the primary oil loop system to the fuel.

The heat exchange system may further comprise an air-oil heat exchanger arranged to transfer heat from the oil of the primary oil loop system.

The air-oil heat exchanger may assist in removing excess heat from the oil of the primary oil loop system that is not transferred to the fuel. This may allow further cooling of the oil beyond the transfer of heat from the oil to the fuel in the fuel-oil heat exchanger.

The air-oil heat exchanger and the primary fuel-oil heat exchanger may be arranged in a parallel configuration on different branches of the primary oil loop system. That is to say, in some implementations, the flow of oil may be split to flow along parallel flow paths such that at least a portion of the oil flows through the primary fuel-oil heat exchanger and at least a portion of the oil flows through another, separate, heat exchanger such as the air-oil heat exchanger and/or an oil-oil heat exchanger, for example. The parallel oil flow-paths may then re-join.

The primary oil loop system may comprise a modulation valve arranged to adjust the oil flow between the branches of the primary oil loop system. This allows variation of the heat transfer ratio between the air-oil and primary fuel-oil heat exchanger. The level of oil flow (and so heat transfer from the oil to the air or fuel) may be modified based on the temperature of the fuel leaving the primary fuel-oil heat exchanger, or entering the pump or combustor (e.g. depending on the location of one or more temperature sensors), and so allow for the control of the fuel temperature, for example within a defined upper and lower bound. This may assist in ensuring improved efficiency of the turbine engine (e.g. by raising the fuel temperature) without risking unnecessary damage to the fuel pump downstream of the fuel-oil heat exchanger by using a temperature that is too high for the durability of the fuel pump.

The primary fuel-oil heat exchanger may be referred to as a main fuel-oil heat exchanger as the majority of the heat transfer between the oil and the fuel may occur in the primary fuel-oil heat exchanger. The primary function of the primary fuel-oil heat exchanger may be to heat the fuel before it is provided to the combustor. At least substantially all of the fuel may pass through the main fuel-oil heat exchanger.

A ratio of the heat transfer from the oil to the fuel for the primary and secondary fuel-oil heat exchangers may be approximately between 70:30 and 90:10, i.e. the primary fuel-oil heat exchanger may be responsible for 70-90% of the heating of the fuel by the oil, with the secondary fuel-oil heat exchanger accounting for the remainder.

A ratio of the heat transfer from the oil to the fuel for the primary and secondary fuel-oil heat exchangers may be approximately 80:20.

In other examples, the proportion of the heat transfer from the oil to fuel due to the secondary fuel-oil heat exchanger may be higher.

The combination of the heat exchange system and the fuel pump may herein be referred to as a fuel management system.

The gas turbine engine may form part of an aircraft. The aircraft may comprise a fuel tank, and a fuel supply pump configured to supply a flow of fuel to the fuel management system of the engine itself. The fuel supply pump may be described as a fuel tank pump or a low pressure pump, and is located upstream of the engine, and therefore upstream of the primary fuel-oil heat exchanger.

The fuel pump of the fuel management system may be described herein as a main fuel pump or engine fuel pump, as, unlike the fuel supply pump, it is a part of the engine itself. The main fuel pump is located downstream of at least one heat exchanger of the heat exchange system—that at least one heat exchanger may be the primary fuel-oil heat exchanger. One or more auxiliary fuel pumps may be located at any suitable position along the fuel flow path.

In some example implementations, a third, fourth or any suitable number of further fuel-oil heat exchangers may be present, optionally with a fuel valve operable to control a flow rate of fuel therethrough.

Oil flow within the gas turbine engine may follow one or more closed loops. The primary oil loop system may be considered a closed loop oil system and may be configured to supply a recirculating flow of oil within the engine and be described as a recirculating lubrication and/or cooling system, or as a recirculating oil system. The primary fuel-oil heat exchanger may be described as forming part of the primary closed loop oil system. The recirculating lubrication and/or cooling system may be described as an oil heat management system, as heat is removed from the oil after it has been warmed in the process of lubricating and/or cooling the gearbox.

The primary oil loop system may comprise at least a first oil pump configured to pump a flow of oil around at least a portion of the recirculating oil system. The first oil pump may be located at any suitable position around the recirculating oil system. The recirculating oil system may be configured such that the flow of oil flows through engine components (e.g. the engine bearing chamber and gearbox) and may then be collected in a sump. The first oil pump may be configured to pump oil from the sump to a first oil tank. As such, the first oil pump may be described as a scavenge pump.

The oil tank may be suitable for containing a volume of oil. The oil tank may be configured to contain any suitable volume of oil. The oil tank may be arranged to remove gasses from the oil in the first oil tank, for example comprising a de-aerator. Oil leaving the oil tank may pass through a filter, strainer, or the like.

A second oil pump may be located between the first oil tank and the primary fuel-oil heat exchanger. The second oil pump may be described as a feed pump. The second oil pump may be configured to pump oil from the first oil tank to the primary fuel-oil heat exchanger and (where present) the air-oil heat exchanger.

At cruise conditions, and for all aspects described herein, the flow of oil entering any of the fuel-oil heat exchangers may have a higher average temperature than the flow of fuel entering the same fuel-oil heat exchanger at cruise conditions. In this way, thermal energy may be transferred from the flow of oil to the flow of fuel flowing through the one or more fuel-oil heat exchangers at cruise conditions. As such,

oil leaving each heat exchanger may have a lower temperature than the oil entering that heat exchanger at cruise conditions.

The secondary oil loop system may have a similar architecture to the primary oil loop system (i.e. an oil tank, one or more pumps), but may supply oil to the integrated drive generator (instead of a main gearbox), and through the secondary fuel-oil heat exchanger (instead of the primary fuel-oil heat exchanger).

Where the primary and/or secondary closed loop systems comprise at least one fuel-oil heat exchanger and at least one air-oil heat exchanger, at least a portion of the flow of oil may not flow through the fuel-oil heat exchanger and/or the air-oil heat exchanger.

For example, when at least one fuel-oil heat exchanger and at least one air-oil heat exchanger are provided in flow series, at least one flow bypass may be configured to allow at least a portion of the flow of oil to bypass, and therefore not flow through, the fuel-oil heat exchanger and/or the air-oil heat exchanger.

When at least one fuel-oil heat exchanger and at least one air-oil heat exchanger are provided in parallel, the recirculating lubrication system may be configured such that any suitable percentage of oil flows through each of the fuel-oil and air-oil heat exchangers. A flow bypass may again be provided by one or each heat exchanger, to allow fuel in that branch of the parallel arrangement to bypass the or each heat exchanger on that branch.

As noted elsewhere herein, the present disclosure may apply to any relevant configuration of gas turbine engine. Such a gas turbine engine may be, for example, a turbofan gas turbine engine, an open rotor gas turbine engine (in which the propeller is not surrounded by a nacelle), a turboprop engine or a turbojet engine. Any such engine may or may not be provided with an afterburner. Such a gas turbine engine may be, for example, configured for land or marine-based power generation applications.

A gas turbine engine in accordance with any aspect of the present disclosure may comprise an engine core comprising a turbine, a combustor, a compressor, and a core shaft connecting the turbine to the compressor. Such a gas turbine engine may comprise a fan (having fan blades). Such a fan may be located upstream of the engine core. Alternatively, in some examples, the gas turbine engine may comprise a fan located downstream of the engine core, for example where the gas turbine engine is an open rotor or a turboprop engine (in which case the fan may be referred to as a propeller).

Where the gas turbine engine is an open rotor or a turboprop engine, the gas turbine engine may comprise two contra-rotating propeller stages attached to and driven by a free power turbine via a shaft. The propellers may rotate in opposite senses so that one rotates clockwise and the other anti-clockwise around the engine's rotational axis. Alternatively, the gas turbine engine may comprise a propeller stage and a guide vane stage configured downstream of the propeller stage. The guide vane stage may be of variable pitch. Accordingly, high pressure, intermediate pressure, and free power turbines respectively may drive high and intermediate pressure compressors and propellers by suitable interconnecting shafts. Thus, the propellers may provide the majority of the propulsive thrust.

Where the gas turbine engine is an open rotor or a turboprop engine, one or more of the propeller stages may be driven by a gearbox. The gearbox may be of the type described herein.

An engine according to the present disclosure may be a turbofan engine. Such an engine may be a direct-drive

turbofan engine in which the fan is directly connected to the fan-drive turbine, for example without a gearbox, via a core shaft. In such a direct-drive turbofan engine, the fan may be said to rotate at the same rotational speed as the fan-drive turbine. Purely by way of example, the fan-drive turbine may be a first turbine, the core shaft may be a first core shaft, and the gas turbine engine may further comprise a second turbine and a second core shaft connecting the second turbine to the compressor. The second turbine, the compressor, and second core shaft may be arranged to rotate at a higher rotational speed than the first core shaft. In such an arrangement, the second turbine may be positioned axially upstream of the first turbine.

An engine according to the present disclosure may be a geared turbofan engine. In such an arrangement, the engine has a fan that is driven via a gearbox. Accordingly, such a gas turbine engine may comprise a gearbox that receives an input from the core shaft and outputs drive to the fan so as to drive the fan at a lower rotational speed than the core shaft. The input to the gearbox may be directly from the core shaft, or indirectly from the core shaft, for example via a spur shaft and/or gear. The core shaft may rigidly connect the turbine and the compressor, such that the turbine and compressor rotate at the same speed (with the fan rotating at a lower speed).

The gas turbine engine as described and/or claimed herein may have any suitable general architecture. For example, the gas turbine engine may have any desired number of shafts that connect turbines and compressors, for example one, two or three shafts. Purely by way of example, the turbine connected to the core shaft may be a first turbine, the compressor connected to the core shaft may be a first compressor, and the core shaft may be a first core shaft. The engine core may further comprise a second turbine, a second compressor, and a second core shaft connecting the second turbine to the second compressor. The second turbine, second compressor, and second core shaft may be arranged to rotate at a higher rotational speed than the first core shaft.

In such an arrangement, the second compressor may be positioned axially downstream of the first compressor. The second compressor may be arranged to receive (for example directly receive, for example via a generally annular duct) flow from the first compressor.

The gearbox may be arranged to be driven by the core shaft that is configured to rotate (for example in use) at the lowest rotational speed (for example the first core shaft in the example above). For example, the gearbox may be arranged to be driven only by the core shaft that is configured to rotate (for example in use) at the lowest rotational speed (for example only by the first core shaft, and not the second core shaft, in the example above). Alternatively, the gearbox may be arranged to be driven by any one or more shafts, for example the first and/or second shafts in the example above.

The gearbox may be a reduction gearbox (in that the output to the fan is a lower rotational rate than the input from the core shaft). Any type of gearbox may be used. For example, the gearbox may be a "planetary" or "star" gearbox, as described in more detail elsewhere herein. Such a gearbox may be a single stage. Alternatively, such a gearbox may be a compound gearbox, for example a compound planetary gearbox (which may have the input on the sun gear and the output on the ring gear, and thus be referred to as a "compound star" gearbox), for example having two stages of reduction.

The gearbox may have any desired reduction ratio (defined as the rotational speed of the input shaft divided by the

11

rotational speed of the output shaft), for example greater than 2.5, for example in the range of from 3 to 4.2, or 3.2 to 3.8, for example on the order of or at least 3, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4, 4.1 or 4.2. The gear ratio may be, for example, between any two of the values in the previous sentence. Purely by way of example, the gearbox may be a “star” gearbox having a reduction ratio in the range of from 3.1 or 3.2 to 3.8. Purely by way of further example, the gearbox may be a “star” gearbox having a reduction ratio in the range 3.0 to 3.1. Purely by way of further example, the gearbox may be a “planetary” gearbox having a reduction ratio in the range 3.6 to 4.2. In some arrangements, the gear ratio may be outside these ranges.

In any gas turbine engine as described and/or claimed herein, fuel of a given composition or blend is provided to a combustor, which may be provided downstream of the fan and compressor(s) with respect to the flowpath (for example axially downstream). For example, the combustor may be directly downstream of (for example at the exit of) the second compressor, where a second compressor is provided. By way of further example, the flow at the exit to the combustor may be provided to the inlet of the second turbine, where a second turbine is provided. The combustor may be provided upstream of the turbine(s).

The or each compressor (for example the first compressor and second compressor as described above) may comprise any number of stages, for example multiple stages. Each stage may comprise a row of rotor blades and a row of stator vanes, which may be variable stator vanes (in that their angle of incidence may be variable). The row of rotor blades and the row of stator vanes may be axially offset from each other. For example, the gas turbine engine may be a direct-drive turbofan gas turbine engine comprising 13 or 14 compressor stages (in addition to the fan). Such an engine may, for example, comprise 3 stages in the first (or “low pressure”) compressor and either 10 or 11 stages in the second (or “high pressure”) compressor. By way of further example, the gas turbine engine may be a “geared” gas turbine engine (in which the fan is driven by a first core shaft via a reduction gearbox) comprising 11, 12 or 13 compressor stages (in addition to the fan). Such an engine may comprise 3 or 4 stages in the first (or “low pressure”) compressor and 8 or 9 stages in the second (or “high pressure”) compressor. By way of further example, the gas turbine engine may be a “geared” gas turbine engine having 4 stages in the first (or “low pressure”) compressor and 10 stages in the second (or “high pressure”) compressor.

The or each turbine (for example the first turbine and second turbine as described above) may comprise any number of stages, for example multiple stages. Each stage may comprise a row of rotor blades and a row of stator vanes, or vice versa, as required. The respective rows of rotor blades and stator vanes may be axially offset from each other. The second (or “high pressure”) turbine may comprise 2 stages in any arrangement (for example regardless of whether it is a geared or direct-drive engine). The gas turbine engine may be a direct-drive gas turbine engine comprising a first (or “low pressure”) turbine having 5, 6 or 7 stages. Alternatively, the gas turbine engine may be a “geared” gas turbine engine comprising a first (or “low pressure”) turbine having 3 or 4 stages.

Each fan blade may be defined as having a radial span extending from a root (or hub) at a radially inner gas-washed location, or 0% span position, to a tip at a 100% span position. The ratio of the radius of the fan blade at the hub to the radius of the fan blade at the tip may be less than (or on the order of) any of: 0.4, 0.39, 0.38, 0.37, 0.36, 0.35, 0.34,

12

0.33, 0.32, 0.31, 0.3, 0.29, 0.28, 0.27, 0.26, or 0.25. The ratio of the radius of the fan blade at the hub to the radius of the fan blade at the tip may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds), for example in the range of from 0.28 to 0.32 or 0.29 to 0.30. These ratios may commonly be referred to as the hub-to-tip ratio. The radius at the hub and the radius at the tip may both be measured at the leading edge (or axially forwardmost) part of the blade. The hub-to-tip ratio refers, of course, to the gas-washed portion of the fan blade, i.e. the portion radially outside any platform.

The radius of the fan may be measured between the engine centreline and the tip of a fan blade at its leading edge. The fan diameter (which may simply be twice the radius of the fan) may be greater than (or on the order of) any of: 140 cm, 170 cm, 180 cm, 190 cm, 200 cm, 210 cm, 220 cm, 230 cm, 240 cm, 250 cm (around 100 inches), 260 cm, 270 cm (around 105 inches), 280 cm (around 110 inches), 290 cm (around 115 inches), 300 cm (around 120 inches), 310 cm, 320 cm (around 125 inches), 330 cm (around 130 inches), 340 cm (around 135 inches), 350 cm, 360 cm (around 140 inches), 370 cm (around 145 inches), 380 cm (around 150 inches), 390 cm (around 155 inches), 400 cm, 410 cm (around 160 inches) or 420 cm (around 165 inches). The fan diameter may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds), for example in the range of from 210 cm to 240 cm, or 250 cm to 280 cm, or 320-cm to 380 cm. Purely by way of non-limitative example, the fan diameter may be in the range of from 170 cm to 180 cm, 190 cm to 200 cm, 200 cm to 210 cm, 210 cm to 230 cm, 290 cm to 300 cm or 340 cm to 360 cm.

The rotational speed of the fan may vary in use. Generally, the rotational speed is lower for fans with a higher diameter. Purely by way of non-limitative example, the rotational speed of the fan at cruise conditions may be less than 3500 rpm, for example less than 2600 rpm, or less than 2500 rpm, or less than 2300 rpm. Purely by way of further non-limitative example, the rotational speed of the fan at cruise conditions for a “geared” gas turbine engine having a fan diameter in the range of from 200 cm to 210 cm may be in the range of from 2750 to 2900 rpm. Purely by way of further non-limitative example, the rotational speed of the fan at cruise conditions for a “geared” gas turbine engine having a fan diameter in the range of from 210 cm to 230 cm may be in the range of from 2500 to 2800 rpm. Purely by way of further non-limitative example, the rotational speed of the fan at cruise conditions for a “geared” gas turbine engine having a fan diameter in the range of from 340 cm to 360 cm may be in the range of from 1500 to 1800 rpm. Purely by way of further non-limitative example, the rotational speed of the fan at cruise conditions for a direct drive engine having a fan diameter in the range of from 190 cm to 200 cm may be in the range of from 3600 to 3900 rpm. Purely by way of further non-limitative example, the rotational speed of the fan at cruise conditions for a direct drive engine having a fan diameter in the range of from 300 cm to 340 cm may be in the range of from 2000 to 2800 rpm.

In use of the gas turbine engine, the fan (with associated fan blades) rotates about a rotational axis. This rotation results in the tip of the fan blade moving with a velocity U_{tip} . The work done by the fan blades on the flow results in an enthalpy rise dH of the flow. A fan tip loading may be defined as dH/U_{tip}^2 , where dH is the enthalpy rise (for example the 1-D average enthalpy rise) across the fan and U_{tip} is the (translational) velocity of the fan tip, for example

at the leading edge of the tip (which may be defined as fan tip radius at leading edge multiplied by angular speed). The fan tip loading at cruise conditions may be greater than (or on the order of) any of: 0.28, 0.29, 0.30, 0.31, 0.32, 0.33, 0.34, 0.35, 0.36, 0.37, 0.38, 0.39 or 0.4 (all values being dimensionless). The fan tip loading may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds), for example in the range of from 0.28 to 0.31, or 0.29 to 0.3 (for example for a geared gas turbine engine).

Gas turbine engines in accordance with the present disclosure may have any desired bypass ratio (BPR), where the bypass ratio is defined as the ratio of the mass flow rate of the flow through the bypass duct to the mass flow rate of the flow through the core. In some arrangements the bypass ratio at cruise conditions may be greater than (or on the order of) any of the following: 9, 9.5, 10, 10.5, 11, 11.5, 12, 12.5, 13, 13.5, 14, 14.5, 15, 15.5, 16, 16.5, 17, 17.5, 18, 18.5, 19, 19.5 or 20. The bypass ratio at cruise conditions may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds), for example in the range of from 12 to 16, 13 to 15, or 13 to 14. Purely by way of non-limitative example, the bypass ratio at cruise conditions of a direct-drive gas turbine engine according to the present disclosure may be in the range of from 9:1 to 11:1. Purely by way of further non-limitative example, the bypass ratio at cruise conditions of a geared gas turbine engine according to the present disclosure may be in the range of from 12:1 to 15:1. The bypass duct may be substantially annular. The bypass duct may be radially outside the core engine. The radially outer surface of the bypass duct may be defined by a nacelle and/or a fan case.

The overall pressure ratio (OPR) of a gas turbine engine as described and/or claimed herein may be defined as the ratio of the stagnation pressure at the exit of the highest pressure compressor (before entry into the combustor) to the stagnation pressure upstream of the fan. By way of non-limitative example, the overall pressure ratio of a gas turbine engine as described and/or claimed herein at cruise conditions may be greater than (or on the order of) any of the following: 35, 40, 45, 50, 55, 60, 65, 70, 75. The overall pressure ratio may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds), for example in the range of from 50 to 70. Purely by way of non-limitative example, the overall pressure ratio at cruise conditions of a geared gas turbine engine having a fan diameter in the range of from 200 cm to 210 cm may be in the range of from 40 to 45. Purely by way of non-limitative example, the overall pressure ratio at cruise conditions of a geared gas turbine engine having a fan diameter in the range of from 210 cm to 230 cm may be in the range of from 45 to 55. Purely by way of non-limitative example, the overall pressure ratio at cruise conditions of a geared gas turbine engine having a fan diameter in the range of from 340 cm to 360 cm may be in the range of from 50 to 60. Purely by way of non-limitative example, the overall pressure ratio at cruise conditions of a direct-drive gas turbine engine having a fan diameter in the range of from 300 cm to 340 cm may be in the range of from 50 to 60.

Specific thrust of an engine may be defined as the net thrust of the engine divided by the total mass flow through the engine. In some examples, specific thrust may depend, for a given thrust condition, upon the specific composition of fuel provided to the combustor. At cruise conditions, the specific thrust of an engine described and/or claimed herein

may be less than (or on the order of) any of the following: 110 Nkg⁻¹ s, 105 Nkg⁻¹ s, 100 Nkg⁻¹ s, 95 Nkg⁻¹ s, 90 Nkg⁻¹ s, 85 Nkg⁻¹ s or 80 Nkg⁻¹ s. The specific thrust may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds), for example in the range of from 80 Nkg⁻¹ s to 100 Nkg⁻¹ s, or 85 Nkg⁻¹ s to 95 Nkg⁻¹ s. Such engines may be particularly efficient in comparison with conventional gas turbine engines. Purely by way of non-limitative example, the specific thrust of a geared gas turbine engine having a fan diameter in the range of from 200 cm to 210 cm may be in the range of from 90 Nkg⁻¹ s to 95 Nkg⁻¹ s. Purely by way of non-limitative example, the specific thrust of a geared gas turbine engine having a fan diameter in the range of from 210 cm to 230 cm may be in the range of from 80 Nkg⁻¹ s to 90 Nkg⁻¹ s. Purely by way of non-limitative example, the specific thrust of a geared gas turbine engine having a fan diameter in the range of from 340 cm to 360 cm may be in the range of from 70 Nkg⁻¹ s to 90 Nkg⁻¹ s. Purely by way of non-limitative example, the specific thrust of a direct drive gas turbine engine having a fan diameter in the range of from 300 cm to 340 cm may be in the range of from 90 Nkg⁻¹ s to 120 Nkg⁻¹ s.

A gas turbine engine as described and/or claimed herein may have any desired maximum thrust. Purely by way of non-limitative example, a gas turbine as described and/or claimed herein may be capable of producing a maximum thrust of at least (or on the order of) any of the following: 100 kN, 110 kN, 120 kN, 130 kN, 135 kN, 140 kN, 145 kN, 150 kN, 155 kN, 160 kN, 170 kN, 180 kN, 190 kN, 200 kN, 250 kN, 300 kN, 350 kN, 400 kN, 450 kN, 500 kN, or 550 kN. The maximum thrust may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds). Purely by way of non-limitative example, a gas turbine as described and/or claimed herein may be capable of producing a maximum thrust in the range of from 155 kN to 170 kN, 330 kN to 420 kN, or 350 kN to 400 kN. Purely by way of non-limitative example, the maximum thrust of a geared gas turbine engine having a fan diameter in the range of from 200 cm to 210 cm may be in the range of from 140 kN to 160 kN. Purely by way of non-limitative example, the maximum thrust of a geared gas turbine engine having a fan diameter in the range of from 210 cm to 230 cm may be in the range of from 150 kN to 200 kN. Purely by way of non-limitative example, the maximum thrust of a geared gas turbine engine having a fan diameter in the range of from 340 cm to 360 cm may be in the range of from 370 kN to 500 kN. Purely by way of non-limitative example, the maximum thrust of a direct drive gas turbine engine having a fan diameter in the range of from 300 cm to 340 cm may be in the range of from 370 kN to 500 kN. The thrust referred to above may be the maximum net thrust at standard atmospheric conditions at sea level plus 15 degrees C. (ambient pressure 101.3 kPa, temperature 30 degrees C.), with the engine static.

In use, the temperature of the flow at the entry to the high pressure turbine may be particularly high. This temperature, which may be referred to as TET, may be measured at the exit to the combustor, for example immediately upstream of the first turbine vane, which itself may be referred to as a nozzle guide vane. In some examples, TET may depend, for a given thrust condition, upon the specific composition of fuel provided to the combustor. At cruise conditions, the TET may be at least (or on the order of) any of the following: 1400K, 1450K, 1500K, 1520K, 1530K, 1540K, 1550K, 1600K or 1650K. Thus, purely by way of non-limitative

example, the TET at cruise conditions of a geared gas turbine engine having a fan diameter in the range of from 200 cm to 210 cm may be in the range of from 1540K to 1600K. Purely by way of non-limitative example, the TET at cruise conditions of a geared gas turbine engine having a fan diameter in the range of from 210 cm to 230 cm may be in the range of from 1590K to 1650K. Purely by way of non-limitative example, the TET at cruise conditions of a geared gas turbine engine having a fan diameter in the range of from 200 cm to 360 cm may be in the range of from 1600K to 1660K. Purely by way of non-limitative example, the TET at cruise conditions of a direct drive gas turbine engine having a fan diameter in the range of from 300 cm to 340 cm may be in the range of from 1590K to 1650K. Purely by way of non-limitative example, the TET at cruise conditions of a direct drive gas turbine engine having a fan diameter in the range of from 300 cm to 340 cm may be in the range of from 1570K to 1630K.

The TET at cruise conditions may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds), for example 1530K to 1600K. The maximum TET in use of the engine may be, for example, at least (or on the order of) any of the following: 1700K, 1750K, 1800K, 1850K, 1900K, 1950K, 2000K, 2050K, or 2100K. Thus, purely by way of non-limitative example, the maximum TET of a geared gas turbine engine having a fan diameter in the range of from 200 cm to 210 cm may be in the range of from 1890K to 1960K. Purely by way of non-limitative example, the maximum TET of a geared gas turbine engine having a fan diameter in the range of from 210 cm to 230 cm may be in the range of from 1890K to 1960K. Purely by way of non-limitative example, the maximum TET of a geared gas turbine engine having a fan diameter in the range of from 340 cm to 360 cm may be in the range of from 1890K to 1960K. Purely by way of non-limitative example, the maximum TET of a direct drive gas turbine engine having a fan diameter in the range of from 300 cm to 340 cm may be in the range of from 1935K to 1995K. Purely by way of non-limitative example, the maximum TET of a direct drive gas turbine engine having a fan diameter in the range of from 300 cm to 340 cm may be in the range of from 1890K to 1950K. The maximum TET may be in an inclusive range bounded by any two of the values in the previous sentence (i.e. the values may form upper or lower bounds), for example in the range of from 1800K to 1950K, or 1900K to 2000K. The maximum TET may occur, for example, at a high thrust condition, for example at a maximum take-off (MTO) condition.

A fan blade and/or aerofoil portion of a fan blade described and/or claimed herein may be manufactured from any suitable material or combination of materials. For example at least a part of the fan blade and/or aerofoil may be manufactured at least in part from a composite, for example a metal matrix composite and/or an organic matrix composite, such as carbon fibre composite. By way of further example at least a part of the fan blade and/or aerofoil may be manufactured at least in part from a metal, such as a titanium based metal or an aluminium based material (such as an aluminium-lithium alloy) or a steel based material. The fan blade may comprise at least two regions manufactured using different materials. For example, the fan blade may have a protective leading edge, which may be manufactured using a material that is better able to resist impact (for example from birds, ice or other material) than the rest of the blade. Such a leading edge may, for example, be manufactured using titanium or a titanium-

based alloy. Thus, purely by way of example, the fan blade may have a carbon-fibre or aluminium based body (such as an aluminium lithium alloy) with a titanium leading edge.

A fan as described and/or claimed herein may comprise a central portion, from which the fan blades may extend, for example in a radial direction. The fan blades may be attached to the central portion in any desired manner. For example, each fan blade may comprise a fixture which may engage a corresponding slot in the hub (or disc). Purely by way of example, such a fixture may be in the form of a dovetail that may slot into and/or engage a corresponding slot in the hub/disc in order to fix the fan blade to the hub/disc. By way of further example, the fan blades may be formed integrally with a central portion. Such an arrangement may be referred to as a bladed disc or a bladed ring. Any suitable method may be used to manufacture such a bladed disc or bladed ring. For example, at least a part of the fan blades may be machined from a block and/or at least part of the fan blades may be attached to the hub/disc by welding, such as linear friction welding.

The gas turbine engines described and/or claimed herein may or may not be provided with a variable area nozzle (VAN). Such a variable area nozzle may allow the exit area of the bypass duct to be varied in use. The general principles of the present disclosure may apply to engines with or without a VAN.

The fan of a gas turbine as described and/or claimed herein may have any desired number of fan blades, for example 14, 16, 18, 20, 22, 24 or 26 fan blades. Where the fan blades have a carbon fibre composite body, there may be 16 or 18 fan blades. Where the fan blades have a metallic body (for example aluminium-lithium or titanium-alloy), there may be 18, 20 or 22 fan blades.

As used herein, the terms idle, taxi, take-off, climb, cruise, descent, approach, and landing have the conventional meaning and would be readily understood by the skilled person. Thus, for a given gas turbine engine for an aircraft, the skilled person would immediately recognise each term to refer to an operating phase of the engine within a given mission of an aircraft to which the gas turbine engine is designed to be attached.

In this regard, ground idle may refer to an operating phase of the engine where the aircraft is stationary and in contact with the ground, but where there is a requirement for the engine to be running. During idle, the engine may be producing between 3% and 9% of the available thrust of the engine. In further non-limitative examples, the engine may be producing between 5% and 8% of available thrust. In further non-limitative examples, the engine may be producing between 6% and 7% of available thrust. Taxi may refer to an operating phase of the engine where the aircraft is being propelled along the ground by the thrust produced by the engine. During taxi, the engine may be producing between 5% and 15% of available thrust. In further non-limitative examples, the engine may be producing between 6% and 12% of available thrust. In further non-limitative examples, the engine may be producing between 7% and 10% of available thrust. Take-off may refer to an operating phase of the engine where the aircraft is being propelled by the thrust produced by the engine. At an initial stage within the take-off phase, the aircraft may be propelled whilst the aircraft is in contact with the ground. At a later stage within the take-off phase, the aircraft may be propelled whilst the aircraft is not in contact with the ground. During take-off, the engine may be producing between 90% and 100% of available thrust. In further non-limitative examples, the engine may be producing between 95% and 100% of available

thrust. In further non-limitative examples, the engine may be producing 100% of available thrust.

Climb may refer to an operating phase of the engine where the aircraft is being propelled by the thrust produced by the engine. During climb, the engine may be producing between 75% and 100% of available thrust. In further non-limitative examples, the engine may be producing between 80% and 95% of available thrust. In further non-limitative examples, the engine may be producing between 85% and 90% of available thrust. In this regard, climb may refer to an operating phase within an aircraft flight cycle between take-off and the arrival at cruise conditions. Additionally or alternatively, climb may refer to a nominal point in an aircraft flight cycle between take-off and landing, where a relative increase in altitude is required, which may require an additional thrust demand of the engine.

As used herein, cruise conditions have the conventional meaning and would be readily understood by the skilled person. Thus, for a given gas turbine engine for an aircraft, the skilled person would immediately recognise cruise conditions to mean the operating point of the engine at mid-cruise of a given mission (which may be referred to in the industry as the “economic mission”) of an aircraft to which the gas turbine engine is designed to be attached. In this regard, mid-cruise is the point in an aircraft flight cycle at which 50% of the total fuel that is burned between top of climb and start of descent has been burned (which may be approximated by the midpoint—in terms of time and/or distance—between top of climb and start of descent. Cruise conditions thus define an operating point of the gas turbine engine that provides a thrust that would ensure steady state operation (i.e. maintaining a constant altitude and constant Mach Number) at mid-cruise of an aircraft to which it is designed to be attached, taking into account the number of engines provided to that aircraft. For example where an engine is designed to be attached to an aircraft that has two engines of the same type, at cruise conditions the engine provides half of the total thrust that would be required for steady state operation of that aircraft at mid-cruise.

In other words, for a given gas turbine engine for an aircraft, cruise conditions are defined as the operating point of the engine that provides a specified thrust (required to provide—in combination with any other engines on the aircraft—steady state operation of the aircraft to which it is designed to be attached at a given mid-cruise Mach Number) at the mid-cruise atmospheric conditions (defined by the International Standard Atmosphere according to ISO 2533 at the mid-cruise altitude). For any given gas turbine engine for an aircraft, the mid-cruise thrust, atmospheric conditions and Mach Number are known, and thus the operating point of the engine at cruise conditions is clearly defined.

Purely by way of example, the forward speed at the cruise condition may be any point in the range of from Mach 0.7 to 0.9, for example 0.75 to 0.85, for example 0.76 to 0.84, for example 0.77 to 0.83, for example 0.78 to 0.82, for example 0.79 to 0.81, for example on the order of Mach 0.8, on the order of Mach 0.85 or in the range of from 0.8 to 0.85. Any single speed within these ranges may be part of the cruise condition. For some aircraft, the cruise conditions may be outside these ranges, for example below Mach 0.7 or above Mach 0.9.

Purely by way of example, the cruise conditions may correspond to standard atmospheric conditions (according to the International Standard Atmosphere, ISA) at an altitude that is in the range of from 10000 m to 15000 m, for example in the range of from 10000 m to 12000 m, for example in the range of from 10400 m to 11600 m (around 38000 ft), for

example in the range of from 10500 m to 11500 m, for example in the range of from 10600 m to 11400 m, for example in the range of from 10700 m (around 35000 ft) to 11300 m, for example in the range of from 10800 m to 11200 m, for example in the range of from 10900 m to 11100 m, for example on the order of 11000 m. The cruise conditions may correspond to standard atmospheric conditions at any given altitude in these ranges.

Purely by way of example, the cruise conditions may correspond to a forward Mach number of 0.8 and standard atmospheric conditions (according to the International Standard Atmosphere) at an altitude of 35000 ft (10668 m). At such cruise conditions, the engine may provide a known required net thrust level. The known required net thrust level is, of course, dependent on the engine and its intended application and may be, for example, a value in the range of from 20 kN to 40 kN.

Purely by way of further example, the cruise conditions may correspond to a forward Mach number of 0.85 and standard atmospheric conditions (according to the International Standard Atmosphere) at an altitude of 38000 ft (11582 m). At such cruise conditions, the engine may provide a known required net thrust level. The known required net thrust level is, of course, dependent on the engine and its intended application and may be, for example, a value in the range of from 35 kN to 65 kN.

In use, a gas turbine engine described and/or claimed herein may operate at the cruise conditions defined elsewhere herein. Such cruise conditions may be determined by the cruise conditions (for example the mid-cruise conditions) of an aircraft to which at least one (for example 2 or 4) gas turbine engine may be mounted in order to provide propulsive thrust.

Furthermore, the skilled person would immediately recognise either or both of descent and approach to refer to an operating phase within an aircraft flight cycle between cruise and landing of the aircraft. During either or both of descent and approach, the engine may be producing between 20% and 50% of available thrust. In further non-limitative examples, the engine may be producing between 25% and 40% of available thrust. In further non-limitative examples, the engine may be producing between 30% and 35% of available thrust. Additionally or alternatively, descent may refer to a nominal point in an aircraft flight cycle between take-off and landing, where a relative decrease in altitude is required, and which may require a reduced thrust demand of the engine.

According to an aspect, there is provided an aircraft comprising a gas turbine engine as described and/or claimed herein. The aircraft according to this aspect is the aircraft for which the gas turbine engine has been designed to be attached. Accordingly, the cruise conditions according to this aspect correspond to the mid-cruise of the aircraft, as defined elsewhere herein.

According to an aspect, there is provided a method of operating a gas turbine engine as described and/or claimed herein. The operation may be at any suitable condition, which may be as defined elsewhere herein (for example in terms of the thrust, atmospheric conditions and Mach Number).

According to an aspect, there is provided a method of operating an aircraft comprising a gas turbine engine as described and/or claimed herein. The operation according to this aspect may include (or may be) operation at any suitable condition, for example the mid-cruise of the aircraft, as defined elsewhere herein.

19

The skilled person will appreciate that except where mutually exclusive, a feature or parameter described in relation to any one of the above aspects may be applied to any other aspect. Furthermore, except where mutually exclusive, any feature or parameter described herein may be applied to any aspect and/or combined with any other feature or parameter described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will now be described by way of example only, with reference to the Figures, in which:

FIG. 1 is a sectional side view of a gas turbine engine;

FIG. 2 is a close-up sectional side view of an upstream portion of a gas turbine engine;

FIG. 3 is a partially cut-away view of a gearbox for a gas turbine engine;

FIG. 4 shows an example aircraft comprising two gas turbine engines;

FIG. 5 is a schematic representation of an example fuel system;

FIG. 6 is a schematic representation of an alternative example fuel system;

FIG. 7 is a schematic representation of a portion of an example recirculating oil system;

FIG. 8 is a schematic representation of a portion of an example recirculating oil system;

FIG. 9 is a schematic representation of a portion of an alternative example recirculating oil system;

FIG. 10 is a schematic representation of a portion of the example fuel system of FIG. 5 and the example recirculating oil system of FIGS. 7 and 8;

FIG. 11 is a schematic representation of a further alternative example recirculating oil system;

FIG. 12 illustrates an example method of operating a gas turbine engine; and

FIG. 13 illustrates a further example method of operating a gas turbine engine.

DETAILED DESCRIPTION OF THE DISCLOSURE

FIG. 1 illustrates a gas turbine engine 10 having a principal rotational axis 9. The engine 10 comprises an air intake 12 and a propulsive fan 23 that generates two airflows: a core airflow A and a bypass airflow B. The gas turbine engine 10 comprises a core 11 that receives the core airflow A. The engine core 11 comprises, in axial flow series, a low pressure compressor 14, a high pressure compressor 15, combustion equipment 16, a high pressure turbine 17, a low pressure turbine 19 and a core exhaust nozzle 20. A nacelle 21 surrounds the gas turbine engine 10 and defines a bypass duct 22 and a bypass exhaust nozzle 18. The bypass airflow B flows through the bypass duct 22. The fan 23 is attached to and driven by the low pressure turbine 19 via a shaft 26 and an epicyclic gearbox 30.

In use, the core airflow A is accelerated and compressed by the low pressure compressor 14 and directed into the high pressure compressor 15 where further compression takes place. The compressed air exhausted from the high pressure compressor 15 is directed into the combustion equipment 16 where it is mixed with fuel F and the mixture is combusted. The combustion equipment 16 may be referred to as the combustor 16, with the terms “combustion equipment 16” and “combustor 16” used interchangeably herein. The resultant hot combustion products then expand through, and thereby drive, the high pressure and low pressure turbines

20

17, 19 before being exhausted through the nozzle 20 to provide some propulsive thrust. The high pressure turbine 17 drives the high pressure compressor 15 by a suitable interconnecting shaft 27. The fan 23 generally acts to impart increased pressure to the bypass airflow B flowing through the bypass duct 22, such that the bypass airflow B is exhausted through the bypass exhaust nozzle 18 to generally provide the majority of the propulsive thrust. The epicyclic gearbox 30 is a reduction gearbox.

An exemplary arrangement for a geared fan gas turbine engine 10 is shown in FIG. 2. The low pressure turbine 19 (see FIG. 1) drives the shaft 26, which is coupled to a sun wheel, or sun gear, 28 of the epicyclic gear arrangement 30. Radially outwardly of the sun gear 28 and intermeshing therewith is a plurality of planet gears 32 that are coupled together by a planet carrier 34. The planet carrier 34 constrains the planet gears 32 to process around the sun gear 28 in synchronicity whilst enabling each planet gear 32 to rotate about its own axis. The planet carrier 34 is coupled via linkages 36 to the fan 23 in order to drive its rotation about the engine axis 9. Radially outwardly of the planet gears 32 and intermeshing therewith is an annulus or ring gear 38 that is coupled, via linkages 40, to a stationary supporting structure 24.

Note that the terms “low pressure turbine” and “low pressure compressor” as used herein may be taken to mean the lowest pressure turbine stages and lowest pressure compressor stages (i.e. not including the fan 23) respectively and/or the turbine and compressor stages that are connected together by the interconnecting shaft 26 with the lowest rotational speed in the engine (i.e. not including the gearbox output shaft that drives the fan 23). In some literature, the “low pressure turbine” and “low pressure compressor” referred to herein may alternatively be known as the “intermediate pressure turbine” and “intermediate pressure compressor”. Where such alternative nomenclature is used, the fan 23 may be referred to as a first, or lowest pressure, compression stage.

The epicyclic gearbox 30 is shown by way of example in greater detail in FIG. 3. Each of the sun gear 28, planet gears 32 and ring gear 38 comprise teeth about their periphery to intermesh with the other gears. However, for clarity only exemplary portions of the teeth are illustrated in FIG. 3. There are four planet gears 32 illustrated, although it will be apparent to the skilled reader that more or fewer planet gears 32 may be provided within the scope of the claimed invention. Practical applications of a planetary epicyclic gearbox 30 generally comprise at least three planet gears 32.

The epicyclic gearbox 30 illustrated by way of example in FIGS. 2 and 3 is of the planetary type, in that the planet carrier 34 is coupled to an output shaft via linkages 36, with the ring gear 38 fixed. However, any other suitable type of epicyclic gearbox 30 may be used. By way of further example, the epicyclic gearbox 30 may be a star arrangement, in which the planet carrier 34 is held fixed, with the ring (or annulus) gear 38 allowed to rotate. In such an arrangement the fan 23 is driven by the ring gear 38. By way of further alternative example, the gearbox 30 may be a differential gearbox in which the ring gear 38 and the planet carrier 34 are both allowed to rotate.

It will be appreciated that the arrangement shown in FIGS. 2 and 3 is by way of example only, and various alternatives are within the scope of the present disclosure. Purely by way of example, any suitable arrangement may be used for locating the gearbox 30 in the engine 10 and/or for connecting the gearbox 30 to the engine 10. By way of further example, the connections (such as the linkages 36, 40

in the FIG. 2 example) between the gearbox 30 and other parts of the engine 10 (such as the input shaft 26, the output shaft and the fixed structure 24) may have any desired degree of stiffness or flexibility. By way of further example, any suitable arrangement of the bearings between rotating and stationary parts of the engine (for example between the input and output shafts from the gearbox and the fixed structures, such as the gearbox casing) may be used, and the disclosure is not limited to the exemplary arrangement of FIG. 2. For example, where the gearbox 30 has a star arrangement (described above), the skilled person would readily understand that the arrangement of output and support linkages and bearing locations would typically be different to that shown by way of example in FIG. 2.

Accordingly, the present disclosure extends to a gas turbine engine having any arrangement of gearbox styles (for example star or planetary), support structures, input and output shaft arrangement, and bearing locations.

Optionally, the gearbox may drive additional and/or alternative components (e.g. the intermediate pressure compressor and/or a booster compressor).

Other gas turbine engines to which the present disclosure may be applied may have alternative configurations. For example, such engines may have an alternative number of compressors and/or turbines and/or an alternative number of interconnecting shafts. By way of further example, the gas turbine engine shown in FIG. 1 has a split flow nozzle 18, 20 meaning that the flow through the bypass duct 22 has its own nozzle 18 that is separate to and radially outside the core engine nozzle 20. However, this is not limiting, and any aspect of the present disclosure may also apply to engines in which the flow through the bypass duct 22 and the flow through the core 11 are mixed, or combined, before (or upstream of) a single nozzle, which may be referred to as a mixed flow nozzle. One or both nozzles (whether mixed or split flow) may have a fixed or variable area.

Whilst the described example relates to a turbofan engine, the disclosure may apply, for example, to any type of gas turbine engine, such as an open rotor (in which the fan stage is not surrounded by a nacelle) or turboprop engine, for example. In some arrangements, the gas turbine engine 10 may not comprise a gearbox 30.

The geometry of the gas turbine engine 10, and components thereof, is defined by a conventional axis system, comprising an axial direction (which is aligned with the rotational axis 9), a radial direction (in the bottom-to-top direction in FIG. 1), and a circumferential direction (perpendicular to the page in the FIG. 1 view). The axial, radial and circumferential directions are mutually perpendicular.

The fuel F provided to the combustion equipment 16 may comprise a fossil-based hydrocarbon fuel, such as Kerosene. Thus, the fuel F may comprise molecules from one or more of the chemical families of n-alkanes, iso-alkanes, cycloalkanes, and aromatics. Additionally or alternatively, the fuel F may comprise renewable hydrocarbons produced from biological or non-biological resources, otherwise known as sustainable aviation fuel (SAF). In each of the provided examples, the fuel F may comprise one or more trace elements including, for example, sulphur, nitrogen, oxygen, inorganics, and metals.

Functional performance of a given composition, or blend of fuel for use in a given mission, may be defined, at least in part, by the ability of the fuel to service the Brayton cycle of the gas turbine engine 10. Parameters defining functional performance may include, for example, specific energy; energy density; thermal stability; and, emissions including particulate matter. A relatively higher specific energy (i.e.

energy per unit mass), expressed as MJ/kg, may at least partially reduce take-off weight, thus potentially providing a relative improvement in fuel efficiency. A relatively higher energy density (i.e. energy per unit volume), expressed as MJ/L, may at least partially reduce take-off fuel volume, which may be particularly important for volume-limited missions or military operations involving refuelling. A relatively higher thermal stability (i.e. inhibition of fuel to degrade or coke under thermal stress) may permit the fuel to sustain elevated temperatures in the engine and fuel injectors, thus potentially providing relative improvements in combustion efficiency. Reduced emissions, including particulate matter, may permit reduced contrail formation, whilst reducing the environmental impact of a given mission. Other properties of the fuel may also be key to functional performance. For example, a relatively lower freeze point ($^{\circ}$ C.) may allow long-range missions to optimise flight profiles; minimum aromatic concentrations (%) may ensure sufficient swelling of certain materials used in the construction of o-rings and seals that have been previously exposed to fuels with high aromatic contents; and, a maximum surface tension (mN/m) may ensure sufficient spray break-up and atomisation of the fuel.

The ratio of the number of hydrogen atoms to the number of carbon atoms in a molecule may influence the specific energy of a given composition, or blend of fuel. Fuels with higher ratios of hydrogen atoms to carbon atoms may have higher specific energies in the absence of bond strain. For example, fossil-based hydrocarbon fuels may comprise molecules with approximately 7 to 18 carbons, with a significant portion of a given composition stemming from molecules with 9 to 15 carbons, with an average of 12 carbons.

A number of sustainable aviation fuel blends have been approved for use. For example, some approved blends comprise blend ratios of up to 10% sustainable aviation fuel, whilst other approved blends comprise blend ratios of between 10% and 50% sustainable aviation fuel (the remainder comprising one or more fossil-based hydrocarbon fuels, such as Kerosene), with further compositions awaiting approval. However, there is an anticipation in the aviation industry that sustainable aviation fuel blends comprising up to (and including) 100% sustainable aviation fuel (SAF) will be eventually approved for use.

Sustainable aviation fuels may comprise one or more of n-alkanes, iso-alkanes, cyclo-alkanes, and aromatics, and may be produced, for example, from one or more of synthesis gas (syngas); lipids (e.g. fats, oils, and greases); sugars; and alcohols. Thus, sustainable aviation fuels may comprise either or both of a lower aromatic and sulphur content, relative to fossil-based hydrocarbon fuels. Additionally or alternatively, sustainable aviation fuels may comprise either or both of a higher iso-alkane and cyclo-alkane content, relative to fossil-based hydrocarbon fuels. Thus, in some examples, sustainable aviation fuels may comprise either or both of a density of between 90% and 98% that of kerosene and a calorific value of between 101% and 105% that of kerosene.

Owing to at least in part to the molecular structure of sustainable aviation fuels, sustainable aviation fuels may provide benefits including, for example, one or more of a higher specific energy (despite, in some examples, a lower energy density); higher specific heat capacity; higher thermal stability; higher lubricity; lower viscosity; lower surface tension; lower freeze point; lower soot emissions; and, lower CO₂ emissions, relative to fossil-based hydrocarbon fuels (e.g. when combusted in the combustion equipment 16). Accordingly, relative to fossil-based hydrocarbon fuels,

such as Kerosene, sustainable aviation fuels may lead to either or both of a relative decrease in specific fuel consumption, and a relative decrease in maintenance costs.

FIG. 4 shows an aircraft 1 with a propulsion system 2 comprising two gas turbine engines 10. The gas turbine engines 10 are supplied with fuel from a fuel supply system onboard the aircraft 1. The fuel supply system of the examples pictured comprises a single fuel source 50, 53.

For the purposes of the present application the term “fuel source” means either 1) a single fuel tank or 2) a plurality of fuel tanks which are fluidly interconnected.

In the present examples, the first (and, in these examples, only) fuel source comprises a centre fuel tank 50, located primarily in the fuselage of the aircraft 1 and a plurality of wing fuel tanks 53a, 53b, where at least one wing fuel tank is located in the port wing and at least one wing fuel tank is located in the starboard wing for balancing. All of the tanks 50, 53 are fluidly interconnected in the example shown, so forming a single fuel source. Each of the centre fuel tank 50 and the wing fuel tanks 53 may comprise a plurality of fluidly interconnected fuel tanks. It will be appreciated that this tank arrangement is provided by way of example only, and is not limiting on the scope of this disclosure. In other examples, an aircraft 1 may have only a single fuel tank 50, and/or the wing fuel tanks 53 may be larger than the central fuel tank 50, or no central fuel tank may be provided (with all fuel instead being stored in the aircraft's wings)—it will be appreciated that many different tank layouts are envisaged and that the examples pictured are provided for ease of description and not intended to be limiting. For example, the wing tanks 53a, 53b may be fluidly interconnected with each other, but fluidly isolated from the centre fuel tank 50, providing two separate fuel sources, which may contain chemically distinct fuels.

An example fuel system 1000 comprising a fuel flow path from the fuel tank 50 (selected as an example tank 50, 53) to the combustor 16 of the gas turbine engine 10 of the aircraft 1 is shown schematically in FIG. 5. Fuel system 1000 comprises both the fuel supply system 50, 1002 (which supplies fuel to the engine) and the fuel management system 1500 (which operates within the engine) of the aircraft 1. The fuel management system 1500 manages fuel temperature as well as fuel flow, directing the fuel via one or more heat exchangers 1004, 1006 of the engine's heat exchange system.

Fuel is pumped from the fuel tank 50 to the gas turbine engine 10 by a low pressure fuel supply pump 1002. The fuel then flows through a secondary fuel-oil heat exchanger 1004 and a primary fuel-oil heat exchanger 1006 before being pumped by an engine fuel pump 1003. The engine fuel pump 1003 may be described as a main fuel pump. The primary fuel-oil heat exchanger 1006 may be described as a main fuel-oil heat exchanger as the oil flowing therethrough may be used to cool and/or lubricate the main gearbox 30 of the engine 10. The secondary fuel-oil heat exchanger 1004 may be described as an integrated drive generator fuel-oil heat exchanger, as the oil flowing therethrough may be used to cool and/or lubricate an integrated drive generator (IDG) of the engine 10. The fuel management system 1500 pictured is arranged so that the fuel reaches the secondary fuel-oil heat exchanger 1004 before it reaches the primary fuel-oil heat exchanger 1006.

The primary fuel-oil heat exchanger 1006 and the secondary fuel-oil heat exchanger 1004 are configured such that a flow of oil is conveyed through each in addition to the flow of fuel therethrough. In the implementation being described, the same fuel flows through both 1004, 1006, but the flows

of oil are different. The primary fuel-oil heat exchanger 1006 and secondary fuel-oil heat exchanger 1004 are configured such that heat may be transferred between the oil and the fuel flowing therethrough. At cruise conditions, the average temperature of the flow of oil entering the primary fuel-oil heat exchanger 1006 and the secondary fuel-oil heat exchanger 1004 is higher than the average temperature of the fuel entering the primary fuel-oil heat exchanger 1006 and the secondary fuel-oil heat exchanger 1004 respectively. In this way, the primary fuel-oil heat exchanger 1006 and the secondary fuel-oil heat exchanger 1004 are configured to transfer thermal energy from a flow of oil to a flow of fuel flowing therethrough at cruise conditions.

The primary fuel-oil heat exchanger 1006 may have oil used to lubricate and/or cool a main gearbox 30 (and optionally also other components 33) of the gas turbine engine 10 passing therethrough, and may therefore be described as a main fuel-oil heat exchanger. The IDG fuel-oil heat exchanger 1004 may have oil used to lubricate and/or cool one or more components of an integrated drive generator 2006 passing therethrough, and may therefore be described as an integrated drive generator fuel-oil heat exchanger.

In general, at least the majority of the fuel which passes through the IDG fuel-oil heat exchanger 1004 also passes through the main fuel-oil heat exchanger 1006, although each heat exchanger 1004, 1006 may be provided with a bypass to allow some of the fuel to avoid passing through the respective heat exchanger for example in the form of a bypass pipe 1005 as shown in FIG. 5. A valve (not shown) may determine what proportion of the fuel passes through the heat exchanger 1006 and what proportion through the bypass pipe 1005.

The two heat exchangers 1004, 1006 may therefore be described as being in series with each other, with respect to fuel flow. The IDG fuel-oil heat exchanger 1004 and the main fuel-oil heat exchanger 1006 are configured such that a flow of oil is also conveyed through each—the oil flowing through one is different from the oil flowing through the other in the implementation being described, although it will be appreciated that the same oil may flow through one fuel-oil heat exchanger and then through another fuel-oil heat exchanger in other implementations. The two heat exchangers 1004, 1006 are therefore in separate closed loop systems 2000', 2000, with respect to oil flow, in the implementation being described, i.e. the oils flowing through the primary and secondary fuel-oil heat exchangers 1006, 1004 may be fluidly separate, and may be chemically distinct from each other.

FIG. 6 shows an alternative example fuel system 6000, comprising a fuel supply system and fuel management system 6500 comprising a fuel flow path from the fuel tank 50 to the combustor 16 of the gas turbine engine 10 of the aircraft 1. As with fuel system 1000, fuel is pumped from the fuel tank 50 by the low pressure fuel supply pump 1002. The fuel then flows through a secondary fuel-oil heat exchanger 1004 and primary fuel-oil heat exchanger 1006 before reaching an engine fuel pump 1003, which pumps the fuel along its flow path. The engine fuel pump 1003 may be described as a main fuel pump.

Fuel system 6000 differs from fuel system 1000 in that fuel system 6000 comprises a recirculation valve 6010 located downstream of the primary fuel-oil heat exchanger 1006 and arranged to recirculate at least a portion of the fuel back to the primary fuel-oil heat exchanger 1006, allowing further heat transfer between the oil of the primary loop system and the fuel.

In the illustrated example, the recirculation valve **6010** is located downstream of the primary fuel-oil heat exchanger **1006**. In the illustrated example, the recirculation valve **6010** is positioned upstream of the engine fuel pump **1003**. The recirculation valve **6010** is arranged to allow a controlled amount of fuel to be returned to the inlet of the primary fuel-oil heat exchanger **1006**, so flowing through the primary fuel-oil heat exchanger **1006** multiple times before reaching the pump **1003**. This recirculation provides a mechanism of controlling the fuel flow of the fuel system **6000**. It is envisaged that the recirculation valve could be positioned downstream of the engine fuel pump **1003**, such that a proportion of the fuel would be recirculated through the heat exchanger **1006** and the pump **1003**, passing through both at least twice. Controlling the fuel flow in the fuel system **6000** can work to provide desired characteristics of the fuel (such as temperature and lubricity) on entry to the combustor **16** or pump **1003**.

A pipe **6011**, which may be referred to as a recirculation pipe as it transports fuel from one point along the main flow-path through the engine to an earlier point along that flow-path so as to allow that fuel to pass through components on that main flow-path multiple times, may therefore be provided, leading from the recirculation valve to a point on the flow-path upstream of the inlet to the primary fuel-oil heat exchanger **1006**. In the illustrated example shown in FIG. 6, the recirculation pipe **6011** returns the recirculated fuel from the heat exchanger outlet, upstream of the pump, to the heat exchanger inlet **1006a**, such that the fuel passes through that heat exchanger **1006** multiple times. In other examples, the recirculation pipe **6011** may take fuel from downstream of both the heat exchanger **1006** and the pump **1003** and return that fuel to a point on the flow-path upstream of both the main pump **1003** and the primary fuel-oil heat exchanger **1006**, such that the recirculated fuel passes through both of those components an additional time. In some implementations, a recirculation pipe **6011** and a bypass pipe **1005** may be provided for any given heat exchanger **1004**, **1006**. In some implementations, the same pipe **1005**, **6011** may serve as both a recirculation pipe and a bypass pipe-one or more valves and/or controllable pumps may be used to control the direction of fluid flow there-through.

The aircraft **1** comprises a recirculating oil system arranged to supply oil to lubricate and remove heat from a plurality of components. The recirculating oil system comprises a primary closed loop oil system. An example of the primary closed loop oil system **2000** is shown schematically in FIG. 7.

The primary closed loop oil system **2000** comprises an oil tank **2002** suitable for containing a volume of oil. In some implementations, gasses are removed from oil within the oil tank **2002** by a de-aerator.

A feed pump **2004** is configured to pump oil from the oil tank **2002** to the main fuel-oil heat exchanger **1006**. The average temperature of the oil entering the main fuel-oil heat exchanger **1006**, at cruise conditions, is higher than the average temperature of fuel entering the main fuel-oil heat exchanger **1006**. In the main fuel-oil heat exchanger **1006**, thermal energy is transferred from the flow of oil to the flow of fuel. In this way, the average temperature of the flow of oil leaving the main fuel-oil heat exchanger **1006** is lower than the average temperature of the flow of oil entering the main fuel-oil heat exchanger **1006**, so it is cooled before re-use as a lubricant and/or coolant, allowing the cooled oil to remove more heat from the system to be lubricated and/or cooled. Also in this way, the average temperature of the fuel

leaving the main fuel-oil heat exchanger **1006** is higher than the average temperature of fuel entering the main fuel-oil heat exchanger **1006**.

The flow of oil is then conveyed to a power gearbox **30**, which may also be described as a main gearbox **30** of the gas turbine engine **10**. The power gearbox **30** is arranged to receive an input from the core shaft and to output drive to the fan via the fan shaft **42** and comprises gears **28**, **32**, **38** and bearings (e.g. journal bearings) which may be lubricated and/or cooled by the oil. The engine **10** may also comprise one or more additional bearings to support the shafts, which may be journal bearings. The oil may additionally be used to lubricate and/or cool the journal bearings, and generally increases in temperature significantly in use at cruise conditions, so assisting with cooling the bearings and gearbox **30** as the flow of oil transports heat away from the bearings and gearbox **30**. The oil may also be used to lubricate one or more further engine components **33**, e.g. an auxiliary gearbox (AGB) and/or one or more bearing chambers. The AGB **33**, also known as an accessory drive, where present, is a gearbox that forms part of the gas turbine engine **10**, although is not a part of the engine's core **11** and does not drive the fan **23**. The AGB instead drives the engine accessories, e.g. fuel pumps, and generally handles large loads. A relatively large amount of heat may therefore be dumped into the oil from the AGB. One or more bearing chambers may be lubricated by the same oil, and may similarly dump heat into the oil. Per unit of oil flowing therethrough, the AGB and bearing chambers may add more heat to the oil than the main gearbox **30** does in many implementations.

The flow of oil may be split into two or more parallel flows, for example one flow through the main gearbox **30** and one flow through the other engine components, or multiple parallel flows through the main gearbox **30** (e.g. via different components of the gearbox) and separate flows through the AGB and the or each bearing chamber **33**.

From the power gearbox **30** (and/or other engine components **33** such as the AGB, where present), the oil collects in a sump **2008**. A scavenge pump **2010** is configured to pump oil from the sump **2008** back into the oil tank **2002**, ready for re-use.

FIG. 8 illustrates the secondary closed oil loop system **2000'**. The closed loop oil system **2000'** comprises a secondary oil tank **2002'** suitable for containing a volume of oil. In some implementations, gasses are removed from oil within the oil tank **2002'** by a de-aerator.

A secondary feed pump **2004'** is configured to pump oil from the secondary oil tank **2002'** to the IDG fuel-oil heat exchanger **1004**. The average temperature of the oil entering the IDG fuel-oil heat exchanger **1004**, at cruise conditions, is higher than the average temperature of fuel entering the IDG fuel-oil heat exchanger **1004**. In the IDG fuel-oil heat exchanger **1004**, thermal energy is transferred from the flow of oil to the flow of fuel. In this way, the average temperature of the flow of oil leaving the IDG fuel-oil heat exchanger **1004** is lower than the average temperature of the flow of oil entering IDG fuel-oil heat exchanger **1004**. Also in this way, the average temperature of the fuel leaving the IDG fuel-oil heat exchanger **1004** is higher than the average temperature of fuel entering the IDG fuel-oil heat exchanger **1004**.

The flow of oil is then conveyed to/back to an integrated drive generator **2006**, where it lubricates and/or cools moving components and is heated up in the process. In some implementations, the oil may be used primarily as a coolant for the IDG **2006**, and may do minimal or no lubrication.

From the integrated drive generator **2006**, the oil collects in a secondary sump **2008'**. A secondary scavenge pump

2010' is configured to pump oil from the secondary sump **2008'** back into the secondary oil tank **2002'**, ready for re-use.

FIG. 9 shows schematically an alternative example section of the primary closed loop oil system **2000** shown in FIG. 7. In this section a flow of oil is pumped by the feed pump **2004** through a valve **2016**. The valve **2016** is operable to divert the flow of oil to both the main fuel-oil heat exchanger **1006** and a first air-oil heat exchanger **2020**, where the first air-oil heat exchanger **2020** is arranged in parallel with the main fuel-oil heat exchanger **1006**. The oil flow path may be described as branching, with the main fuel-oil heat exchanger **1006** on one branch and the first air-oil heat exchanger **2020** on the other branch, in a parallel configuration such that oil can only flow via one or the other, not both. The flows of oil are then recombined and conveyed to the power gearbox **30** and/or other engine components **33**. Any suitable percentage of oil may flow through each of the first air-oil heat exchanger **2020** and the main fuel-oil heat exchanger **1006**. In some examples, the valve **2016** is operable to vary the flow of oil to the main fuel-oil heat exchanger **1006** and the first air-oil heat exchanger **2020**. In various examples, an oil-oil heat exchanger (not shown) may be arranged in series with the first air-oil heat exchanger **2020**. The oil-oil heat exchanger may allow for heat exchange between the primary and secondary closed loop oil systems.

FIG. 10 shows schematically an example arrangement and interaction of the primary closed loop oil system **2000**, secondary closed loop oil system **2000'** and fuel system **1000**, with fuel flow shown in thick black lines and oil flow in thinner black lines.

The primary closed loop oil system **2000** of this example arrangement is arranged as shown in FIG. 7. The secondary closed loop oil system **2000'** of this example arrangement is arranged as shown in FIG. 8. The fuel system **1000** of this example arrangement is arranged as shown in FIG. 5 with the exception of bypass pipe **1005** and fuel management system **1500**, which are excluded for clarity.

In use, fuel is pumped from the fuel tank **50** by the low pressure fuel pump **1002**. The fuel then flows through the IDG fuel-oil heat exchanger **1004**. The secondary closed loop oil system **2000'** is configured such that its recirculating flow of oil also flows through the IDG fuel-oil heat exchanger **1004**.

At cruise conditions, the average temperature of the flow of oil entering the IDG fuel-oil heat exchanger **1004** is higher than the average temperature of the flow of fuel entering the IDG fuel-oil heat exchanger **1004**. The IDG fuel-oil heat exchanger **1004** is configured such that heat is transferred from the flow of oil to the flow of fuel. In this way, the average temperature of the flow of oil on exit from the IDG fuel-oil heat exchanger **1004** is lower than the average temperature of the flow of oil on entry to the IDG fuel-oil heat exchanger **1004**. In the same way, the average temperature of the flow of fuel on exit from the IDG fuel-oil heat exchanger **1004** is higher than the average temperature of the flow of fuel on entry from the IDG fuel-oil heat exchanger **1004**.

The fuel then flows through the main fuel-oil heat exchanger **1006**. The primary closed loop oil system **2000** is configured such that its recirculating flow of oil also flows through the main fuel-oil heat exchanger **1006**.

At cruise conditions, the average temperature of the flow of oil entering the main fuel-oil heat exchanger **1006** is higher than the average temperature of the flow of fuel entering the main fuel-oil heat exchanger **1006**. The main

fuel-oil heat exchanger **1006** is configured such that heat is transferred from the flow of oil to the flow of fuel. In this way, the average temperature of the flow of oil on exit from the main fuel-oil heat exchanger **1006** is lower than the average temperature of the flow of oil on entry to the main fuel-oil heat exchanger **1006**. In the same way, the average temperature of the flow of fuel on exit from the main fuel-oil heat exchanger **1006** is higher than the average temperature of the fuel on entry to the main fuel-oil heat exchanger **1006**.

At cruise conditions, the average temperature of the flow of oil flowing through the IDG fuel-oil heat exchanger **1004** may be lower than the average temperature of the flow of oil flowing through the main fuel-oil heat exchanger **1006**. In this way, the fuel passes through the heat exchanger having a lower average oil flow temperature first, before passing through the heat exchanger having a higher average oil flow temperature.

After flowing through the main fuel-oil heat exchanger **1006**, the fuel flows to the engine fuel pump **1003**, which is located downstream of the primary and secondary fuel-oil heat exchangers **1006**, **1004** and is arranged to deliver fuel to the combustor **16** of the gas turbine engine **10**.

FIG. 11 shows schematically an example configuration of the primary closed loop oil system **2000** and the secondary closed loop oil system **2000'** where the two independent recirculating flows of oil are brought into a heat exchange relationship through an oil-oil heat exchanger **2030**.

In the example shown in FIG. 11, the primary closed loop oil system **2000** is configured such that the recirculating flow of oil is pumped by the feed pump **2004** through a valve **2016**, which may be referred to as a modulation valve, or primary modulation valve. The primary modulation valve **2016** is operable to divert at least a portion of the flow of oil to each of the main fuel-oil heat exchanger **1006** and a first air-oil heat exchanger **2020**, where the first air-oil heat exchanger **2020** is in series with the oil-oil heat exchanger **2030**, and the air-oil **2020** and oil-oil heat exchanger **2030** arrangement is in parallel with the main fuel-oil heat exchanger **1006**.

In examples, any suitable portion of the flow of oil may be diverted between the main fuel-oil heat exchanger **1006** and the first air-oil heat exchanger **2020**. In examples, the primary modulation valve **2016** is operable to divert a fixed portion of the flow of oil to each of the main fuel-oil heat exchanger **1006** and the first air-oil heat exchanger **2020**. In other examples, the valve **2016** is operable to divert a variable portion of the flow of oil to each of the main fuel-oil heat exchanger **1006** and the first air-oil heat exchanger **2020**, for example using data from a temperature sensor for feedback to control the variable portion.

After flowing through the heat exchangers, the flow of oil in the primary closed loop oil system is then recombined and conveyed to the power gearbox **30** (and/or the other engine components **33**) and then to the sump **2008**. The scavenge pump **2010** then pumps the oil from the sump **2008** to the oil tank **2002**, for re-use.

The flow of oil within the secondary closed loop oil system **2000'** is arranged to be brought into a heat exchange relationship with the separate flow of oil within the primary closed loop oil system **2000** through the oil-oil heat exchanger **2030**. In the oil-oil heat exchanger **2030** the flow of oil within the primary closed loop oil system **2000** does not mix with the flow of oil within the secondary closed loop oil system **2000'**. The oil-oil heat exchanger **2030** is configured such that a transfer of heat may take place between the two separated flows of oil. In this way, heat from a hotter

flow of oil may be transferred to the cooler flow of oil within the oil-oil heat exchanger **2030**.

The secondary closed loop oil system **2000'** is configured such that the recirculating flow of oil is pumped by the secondary feed pump **2004'** through a valve **2016'**, which may be referred to as a secondary modulation valve. The secondary modulation valve **2016'** is operable to divert at least a portion of the flow of oil between the IDG fuel-oil heat exchanger **1004** and the oil-oil heat exchanger **2030**, where the oil-oil heat exchanger **2030** is arranged in parallel with the IDG fuel-oil heat exchanger **1004**.

In examples, any suitable portion of the flow of oil may be diverted between the IDG fuel-oil heat exchanger **1004** and the oil-oil heat exchanger **2030**. In examples, the valve **2016'** is operable to divert a fixed portion of the flow of oil to each of the IDG fuel-oil heat exchanger **1004** and the oil-oil heat exchanger **2030**. In other examples, the secondary modulation valve **2016'** is operable to divert a variable portion of the flow of oil to each of the IDG fuel-oil heat exchanger **1004** and the oil-oil heat exchanger **2030**, for example using data from a temperature sensor for feedback to control the variable portion.

After flowing through the heat exchangers, the flow of oil is then conveyed to the integrated drive generator **2006** and then to the secondary sump **2008'**. The secondary scavenge pump **2010** then pumps the oil from the secondary sump **2008** to the secondary oil tank **2002**, for re-use.

One or more temperature sensors may therefore be provided, for example arranged to sense fuel temperature on approach to the fuel pump **1003** or the combustor **16**. Data from the one or more sensors may be used by a controller of the heat exchange system to manage fuel and/or oil flows through and around the heat exchangers.

The inventors appreciated that careful selection and control of fuel based on lubricity can affect combustion efficiency, in particular with respect to fuel nozzle spray performance within the combustor. Fuel nozzle spray performance affects burn efficiency of the fuel, so engine efficiency can be improved with the selection of desired lubricity. Further, careful control of fuel lubricity can also improve pump performance, and potentially improve pump longevity. For example, using a fuel with a higher lubricity may better-lubricate pump components, and potentially reduce wear on the pump.

FIG. 12 illustrates an example method **12000** of operating a gas turbine engine **10**. The method **12000** comprises the steps of:

Step **12100**: Obtaining a fuel, wherein the lubricity of the fuel to be provided to the gas turbine engine **10** is at least 0.71 mm WSD at 25° C. (ASTM D5001 BOCLE test).

Obtaining the fuel may comprise selecting a single fuel. Obtaining the fuel may alternatively comprise selecting a fuel mixture. The selected fuel may be the only fuel on board of the aircraft **1**. As such, the selection of the fuel may be performed during refuelling of the aircraft **1**. Alternatively, the selected fuel may be one of several fuels on board the aircraft **1**, or a mixture of multiple fuels onboard the aircraft (if multiple separate fuel sources **50**, **53** are available). As such, the selection of the fuel may be performed during flight. The selection process may be automated. For example, an onboard controller may be configured to select and/or mix fuel(s) stored in one or more fuel tanks **50**, **53** in order to provide a fuel with the desired lubricity. The controller may store the standard lubricity values (e.g. values obtained using the standard test procedures in laboratory conditions) and tank locations for the fuels onboard

and calculate the ratio of fuel mixing (which may include no mixing of fuels) in order to provide a fuel with the desired lubricity. The same controller, or another controller, may also control the heat exchange system so as to adjust the lubricity of the selected fuel before it reaches the pump **1003** or combustor **16**, as discussed below.

Step **12200**: Operating the gas turbine engine using the fuel having a lubricity of between 0.71 mm and 0.90 mm wear scar diameter (WSD) at 25° C. (ASTM D5001 BOCLE test). This operating of the engine **10** may comprise delivering the fuel to the combustor **16** of the gas turbine engine **10** via a fuel management system **1500**, **6500**.

As discussed with respect to FIG. 6, the fuel management system **6500** may comprise a primary fuel-oil heat exchanger **1006** and a secondary fuel-oil heat exchanger **1004** arranged to transfer heat between the oil of the primary and secondary oil loop systems **2000** and **2000'** and the fuel. The fuel management system **6500** is arranged so that the fuel reaches the secondary fuel-oil heat exchanger **1004** before the primary fuel-oil heat exchanger **1006**. An engine fuel pump **1003** is positioned downstream of the primary fuel-oil-heat exchanger **1004**. The engine fuel pump **1003** is arranged to deliver the fuel to the combustor **16**.

Optionally, the method may further comprise:

Step **12300**: Controlling the heat exchange system so as to raise the fuel temperature to control the lubricity of the fuel on entry to the pump **1003** and/or combustor **16** at cruise conditions.

The controlling **12300** the heat exchange system may comprise controlling fuel flow through one or both of the primary and secondary fuel-oil heat exchangers **1006**, **1004**.

Fuel flow may be controlled using a recirculation valve **6010** where present, as in fuel management system **6500**. An actively controlled amount of fuel leaving the primary fuel-oil heat exchanger **1006** may be recirculated to the primary fuel-oil heat exchanger **1006** rather than flowing directly to the combustor **16**. This recirculation may also take fuel which has already passed through the engine fuel pump **1003** back to a position upstream of the engine fuel pump **1003**.

Alternatively or additionally, fuel flow may be controlled by using one or more bypass pipes **1005** as shown in FIG. 5, arranged to allow a proportion of the fuel to avoid passing through either or both fuel-oil heat exchangers.

The speed of the pump **1003** may also be adjusted, either speeding up fuel flow rate (so reducing heat transfer per unit volume passing through the heat exchangers) or reducing fuel flow rate (so increasing heat transfer per unit volume passing through the heat exchangers).

The controlling **12300** the heat exchange system may comprise controlling oil flow through the primary and secondary fuel-oil heat exchangers **1006**, **1004**, and/or through one or more other heat exchangers—e.g. an oil-oil heat exchanger **2030** between separate closed loop oil systems or an air-oil heat exchanger **2020**.

Additionally or alternatively, as for fuel flow, oil flow may be controlled using one or more bypass pipes or recirculation pipes, where present, allowing oil to bypass one or more heat exchangers **1004**, **1006** instead of flowing therethrough, or to be recirculated through one or more heat exchangers. Oil may also be recirculated in some implementations, and/or oil flow rate may be adjusted by controlling one or more oil pumps. One or more controllable valves, pumps, and/or sensors may be provided accordingly to facilitate oil flow control. The speed of a pump **2004**, **2010**, **2004'**, **2010'** may also be adjusted, either speeding up oil flow rate or

31

reducing oil flow rate (so increasing heat transfer per unit volume passing through the heat exchangers).

The heat exchange system may comprise a controller arranged to implement this control (of fuel flow and/or oil flow). The controller may receive inputs from one or more temperature sensors, and may control one or more valves (e.g. (e.g. **2016**, **2016'** as shown in FIG. **11** and/or recirculation valve **6010**), and/or the pump **1003**, based on the received data. One or more oil feed pumps and/or scavenge pumps may also be controlled by the controller.

This active control may be performed based on one or more parameters such as:

- Core shaft speed and engine thrust demand;
- Current fuel temperature and/or oil temperature at one or more locations;
- Fuel calorific value;
- Fan rotational speed;
- Fuel flow rate to the combustor (commonly referred to as WFE—Weight of Main Engine Fuel Flow);
- Fuel lubricity; and
- Main/engine fuel pump speed, or speed options.

In alternative examples, the control of the amount of fuel leaving the primary fuel-oil heat exchanger **1006** to be recirculated may not be an active method step—instead, a set, fixed, proportion of the fuel may be recirculated. Alternatively, no fuel may be recirculated, and no recirculation route may be available in some implementations.

The inventors also appreciated that careful control of fuel temperature might allow more efficient use to be made of newer fuels, adjusting parameters at cruise conditions to improve combustion efficiency and/or improve component longevity.

FIG. **13** illustrates an example method **13000** of operating a gas turbine engine **10**. The method **13000** comprises the steps of:

Step **13100**: Providing a fuel to the combustor **16** of the gas turbine engine **10** via a fuel management system **1500**.

The gas turbine engine **10** comprises a heat exchange system comprising at least one fuel-oil heat exchanger, and optionally at least two heat exchangers, or more, as described above.

Step **13200**: Controlling the heat exchange system so as to raise the fuel temperature to at least 135° C. on entry to the combustor at cruise conditions.

The controlling the heat exchange system may comprise controlling fuel flow through the primary and secondary fuel-oil heat exchangers **1006**, **1004**.

Fuel flow may be controlled using the recirculation valve **6010**, as discussed with respect to step **12300** of method **12000**. Alternatively or additionally, fuel flow may be controlled by using one or more bypass pipes **1005** arranged to allow a proportion of the fuel to avoid passing through either or both fuel-oil heat exchangers, as described above with respect to FIG. **12**.

Alternatively or additionally, the controlling the heat exchange system may comprise controlling oil flow through one or more heat exchangers—for example diverting less oil to an air-oil heat exchanger **2020** when the fuel temperature needs raising and diverting more oil to the air-oil heat exchanger **2020** (so allowing more heat to be lost from the engine to the environment) when the fuel temperature needs lowering. Similarly, oil may be diverted past a fuel-oil heat exchanger, e.g. using a bypass pipe, rather than through that heat exchanger if or when less heat input to the fuel is needed.

32

One or more temperature sensors may be provided, and optionally used in an automated feedback process to control the oil and/or fuel flows.

One or more valves may be provided, optionally automatically controlled in a feedback process using data from the temperature sensor(s), to control the oil and/or fuel flows.

Heating the fuel to higher temperatures than previously used may improve the cooling of the oil before it is returned to the rest of the turbine engine, and/or may improve combustion efficiency of the fuel.

The method of FIG. **13** may be used in conjunction with the method of FIG. **12**.

It will be understood that the invention is not limited to the embodiments above-described and various modifications and improvements can be made without departing from the concepts described herein. Except where mutually exclusive, any of the features may be employed separately or in combination with any other features and the disclosure extends to and includes all combinations and sub-combinations of one or more features described herein.

We claim:

1. A method of operating an aircraft propulsion system, the aircraft propulsion system comprising:

a fuel, wherein the fuel includes sustainable aviation fuel (SAF) in an amount up to and including 100% SAF; and

a gas turbine engine comprising:

an engine core comprising a turbine, a compressor, a combustor arranged to combust the fuel, and a core shaft connecting the turbine to the compressor;

a fan located upstream of the engine core;

a fan shaft;

a gearbox that receives an input from the core shaft and outputs drive to the fan via the fan shaft;

a heat exchange system comprising at least one fuel-oil heat exchanger arranged to transfer heat to the fuel;

a fuel pump arranged to deliver the fuel to the combustor, wherein the fuel pump is located downstream of the at least one fuel-oil heat exchanger;

a primary oil loop system arranged to supply oil to the gearbox, wherein the at least one fuel-oil heat exchanger comprises at least a primary fuel-oil heat exchanger arranged such that the oil of the primary oil loop system and fuel flows therethrough to transfer heat from the oil to the fuel;

an integrated drive generator;

a secondary oil loop system arranged to provide oil to the integrated drive generator,

wherein the heat exchange system further comprises an oil-oil heat exchanger arranged to transfer heat between the oil of the primary oil loop system and the oil of the secondary oil loop system; and

a controller configured to control respective fuel and oil flows through the primary fuel-oil heat exchanger and the oil-oil heat exchanger of the heat exchange system, based on a fuel calorific value, to heat the fuel so as to raise the fuel temperature to at least 135° C. on entry to the combustor at cruise conditions;

the method comprising:

operating the aircraft propulsion system at cruise conditions; and

controlling the respective fuel and oil flows through the primary fuel-oil heat exchanger and the oil-oil heat exchanger of the heat exchange system, based on a fuel calorific value, to raise the fuel temperature to at

33

least 135° C. on entry to the combustor while operating the aircraft propulsion system at cruise conditions.

2. The method of claim 1, wherein the fuel temperature is raised to at least 150° C. on entry to the combustor at cruise conditions. 5

3. The method of claim 1, wherein the fuel temperature is raised to at least 170° C. on entry to the combustor at cruise conditions.

4. The method of claim 1, wherein the fuel temperature is raised to at least 190° C. on entry to the combustor at cruise conditions. 10

5. The method of claim 1, wherein the at least one fuel-oil heat exchanger of the heat exchange system further comprises: 15

a secondary fuel-oil heat exchanger arranged to receive the fuel and oil from the secondary oil loop system; and the method further comprises transferring heat between the oil from the secondary oil loop system and the fuel using the secondary fuel-oil heat exchanger.

6. The method of claim 5, wherein:

the fuel flows through the secondary fuel-oil heat exchanger prior to flowing through the primary fuel-oil heat exchanger, such that heat is transferred from the oil in the secondary oil loop system to the fuel before heat is transferred from the oil in the primary oil loop system to the fuel. 25

7. The method of claim 1, wherein:

the heat exchange system further comprises an air-oil heat exchanger arranged to transfer heat from the oil of the primary oil loop system; and 30

the air-oil heat exchanger and the primary fuel-oil heat exchanger are arranged in a parallel configuration on different branches of the primary oil loop system; and wherein the method further comprises adjusting an amount of oil sent to the air-oil heat exchanger based on fuel temperature. 35

8. The method of claim 5, wherein a ratio of heat transfer to the fuel between the primary and secondary fuel-oil heat exchangers is between 70:30 and 90:10. 40

9. The method of claim 1, further comprising controlling an amount of fuel sent to the secondary fuel-oil heat exchanger.

10. The method of claim 1, wherein the heat exchange system comprises at least one bypass pipe arranged to allow fuel to bypass a heat exchanger of the heat exchange system, and wherein the method comprises adjusting an amount of fuel sent through the bypass pipe based on fuel temperature. 45

11. An aircraft propulsion system, comprising:

a fuel, wherein the fuel includes sustainable aviation fuel (SAF) in an amount up to and including 100% SAF; and 50

a gas turbine engine comprising:

an engine core comprising a turbine, a compressor, a combustor arranged to combust the fuel, and a core shaft connecting the turbine to the compressor; 55

a fan located upstream of the engine core;

a fan shaft;

a gearbox arranged to receive an input from the core shaft and output drive to the fan via the fan shaft; 60

a heat exchange system comprising at least one fuel-oil heat exchanger arranged to transfer heat to the fuel;

a fuel pump arranged to deliver the fuel to the combustor, wherein the fuel pump is located downstream of the at least one fuel-oil heat exchanger; 65

a primary oil loop system arranged to supply oil to the gearbox, wherein the at least one fuel-oil heat

34

exchanger comprises at least a primary fuel-oil heat exchanger arranged such that the oil of the primary oil loop system and fuel flows therethrough to transfer heat from the oil to the fuel;

an integrated drive generator;

a secondary oil loop system arranged to provide oil to the integrated drive generator;

wherein the heat exchange system further comprises an oil-oil heat exchanger arranged to transfer heat between the oil of the primary oil loop system and the oil of the secondary oil loop system; and

a controller configured to control respective fuel and oil flows through the primary fuel-oil heat exchanger and the oil-oil heat exchanger of the heat exchange system, based on a fuel calorific value, to heat the fuel so as to raise the fuel temperature to at least 135° C. on entry to the combustor at cruise conditions.

12. The aircraft propulsion system of claim 11, wherein the controller is configured to control the respective fuel and oil flows through the primary fuel-oil heat exchanger and the oil-oil heat exchanger to raise the fuel temperature to at least 150° C. on entry to the combustor at cruise conditions. 20

13. The aircraft propulsion system of claim 11, wherein the controller is configured to control the respective fuel and oil flows through the primary fuel-oil heat exchanger and the oil-oil heat exchanger to raise the fuel temperature to at least 170° C. on entry to the combustor at cruise conditions. 25

14. The aircraft propulsion system of claim 11, wherein the at least one fuel-oil heat exchanger further comprises a secondary fuel-oil heat exchanger arranged such that oil from the secondary oil loop system and the fuel flows therethrough to transfer heat between the oil from the secondary oil loop system and the fuel. 30

15. The aircraft propulsion system of claim 14, wherein the primary fuel-oil heat exchanger and the secondary fuel-oil heat exchanger are arranged such that the fuel flows through the secondary fuel-oil heat exchanger prior to flowing through the primary fuel-oil heat exchanger, such that heat is transferred from the oil in the secondary oil loop system to the fuel before heat is transferred from the oil in the primary oil loop system to the fuel. 35

16. The aircraft propulsion system of claim 11, wherein: the heat exchange system further comprises an air-oil heat exchanger arranged to transfer heat from the oil of the primary oil loop system; and 40

the air-oil heat exchanger and the primary fuel-oil heat exchanger are arranged in a parallel configuration on different branches of the primary oil loop system.

17. The aircraft propulsion system of claim 15, wherein the controller is further configured to control fuel flow through the secondary fuel-oil heat exchanger and through a fuel recirculation valve. 45

18. The aircraft propulsion system of claim 15, wherein the controller is further configured to control fuel flow through at least one bypass pipe so as to avoid passing the fuel flow through at least one of the primary fuel-oil heat exchanger and the secondary fuel-oil heat exchanger. 50

19. The aircraft propulsion system of claim 14, wherein a ratio of the heat transfer from the oil to the fuel for the primary fuel-oil heat exchanger and the secondary fuel-oil heat exchanger is between 70:30 and 90:10.

20. An aircraft propulsion system, comprising:

a gas turbine engine comprising:

an engine core comprising a turbine, a compressor, a combustor arranged to combust a fuel, and a core shaft connecting the turbine to the compressor;

a fan located upstream of the engine core;

35

a fan shaft;
a gearbox arranged to receive an input from the core shaft and output drive to the fan via the fan shaft;
a heat exchange system comprising at least one fuel-oil heat exchanger arranged to transfer heat to the fuel; and
a fuel pump arranged to deliver the fuel to the combustor, wherein the fuel pump is located downstream of the at least one fuel-oil heat exchanger;
a controller configured to control the heat exchange system; and
the fuel, wherein the heat exchange system includes:
a primary oil loop system arranged to supply oil to the gearbox, wherein the at least one fuel-oil heat exchanger comprises at least a primary fuel-oil heat exchanger and a secondary fuel-oil heat exchanger, the primary fuel-oil heat exchanger being arranged such that the oil of the primary oil loop system and fuel flows therethrough to transfer heat from the oil to the fuel;
an integrated drive generator; and
a secondary oil loop system arranged to provide oil to the integrated drive generator;

36

wherein the secondary fuel-oil heat exchanger is arranged such that oil from the secondary oil loop system and the fuel flows therethrough, to transfer heat between the oil from the secondary oil loop system and the fuel, and the primary oil loop system, the secondary oil loop system, the primary fuel-oil heat exchanger and the secondary fuel-oil heat exchanger are arranged such that the fuel flows through the secondary fuel-oil heat exchanger prior to flowing through the primary fuel-oil heat exchanger, to transfer heat from the oil in the secondary oil loop system to the fuel before heat is transferred from the oil in the primary oil loop system to the fuel,
wherein the controller is configured to control fuel flow through each of the primary fuel-oil heat exchanger and the secondary fuel-oil heat exchanger of the heat exchange system based on a fuel calorific value to heat the fuel so as to raise the fuel temperature to at least 135° C. on entry to the combustor at cruise conditions, and
the fuel includes sustainable aviation fuel (SAF) in an amount up to and including 100% SAF.

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